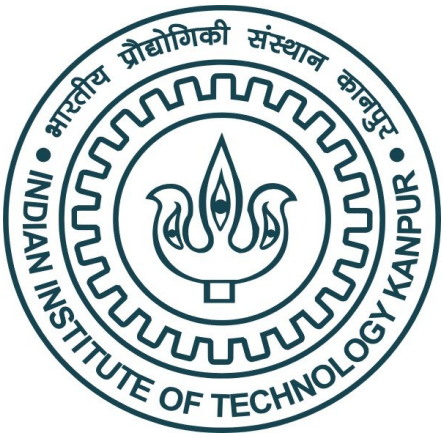
Some Topics in Model Theory: Ultrafilters (MTH496)

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Abstract

Two structures are elementarily equivalent if they satisfy the same first-order sentences. The Keisler-Shelah theorem characterizes this syntactic notion algebraically: $\mathcal{M} \equiv \mathcal{N}$ if and only if $\mathcal{M}$ and $\mathcal{N}$ have isomorphic ultrapowers with respect to a common ultrafilter. This report develops the theory of ultrafilters and ultraproducts needed for Keisler's proof of this theorem under the Generalized Continuum Hypothesis (GCH). We begin with filters and ultrafilters, show that there are 2^{2^κ} ultrafilters on a set of infinite cardinality κ, and prove Łoś' theorem along with some applications, such as the Compactness Theorem and ultraproducts of ultraproducts via product ultrafilters. After recalling saturated and universal models, we study countably incomplete ultrafilters, the Rudin-Keisler ordering, and regular ultrafilters, whose ultrapowers are highly universal. We then introduce good ultrafilters, show that for countably incomplete ultrafilters goodness is equivalent to saturation of all ultrapowers, and construct κ^+-good ultrafilters on κ. Under GCH, the resulting ultrapowers are saturated of cardinality κ^+, and the uniqueness of saturated models yields the theorem.

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1 Introduction

Two $\mathcal{L}$-structures $\mathcal{M}$ and $\mathcal{N}$ are *elementarily equivalent*, written $\mathcal{M} \equiv \mathcal{N}$, if they satisfy the same $\mathcal{L}$-sentences. Isomorphic structures are clearly elementarily equivalent, but the converse fails. For instance, $(\mathbb{Z}, <)$ and the lexicographic sum $(\mathbb{Z} + \mathbb{Z}, <)$ are elementarily equivalent but not isomorphic. Elementary equivalence is a syntactic notion, since it refers to sentences, while isomorphism is an algebraic one. This report is about a theorem that relates the two.

The relevant construction is that of an ultraproduct. Given a family of structures $\{\mathcal{M}_i : i \in I\}$ and an ultrafilter $\mathcal{U}$ over I, the ultraproduct $\prod_{\mathcal{U}} \mathcal{M}_i$ is obtained by identifying two elements of $\prod_I M_i$ if they agree on a "large" set of indices, i.e., one that belongs to $\mathcal{U}$. Łoś' theorem says that a first-order sentence holds in the ultraproduct if and only if it holds in a large set of the factors. In particular, every structure is elementarily equivalent to each of its ultrapowers. So, if $\mathcal{M}$ and $\mathcal{N}$ have isomorphic ultrapowers with respect to the same ultrafilter $\mathcal{U}$, then

$$\mathcal{M} \equiv \prod_{\mathcal{U}} \mathcal{M} \cong \prod_{\mathcal{U}} \mathcal{N} \equiv \mathcal{N}.$$

Remarkably, the converse holds as well.

The Keisler-Shelah Theorem

Let $\mathcal{M}, \mathcal{N}$ be $\mathcal{L}$-structures. Then, $\mathcal{M} \equiv \mathcal{N}$ if and only if there are an index set I and an ultrafilter $\mathcal{U}$ over I such that $\prod_{\mathcal{U}} \mathcal{M} \cong \prod_{\mathcal{U}} \mathcal{N}$.

So, elementary equivalence admits a description that does not mention formulae at all. Keisler proved this theorem in 1961 assuming the Generalized Continuum Hypothesis (GCH). Shelah removed the hypothesis in 1971 using a different argument, based on iterated ultrapowers, which we do not discuss. The aim of this report is to develop the theory of ultrafilters and ultraproducts needed for Keisler's proof, and to end with a proof of the Keisler-Shelah theorem under GCH.

Let us outline this proof, as it explains the choice of topics. Assume GCH and $\mathcal{M} \equiv \mathcal{N}$. Fix an infinite cardinal $\lambda \geq |\mathcal{L}|, |M|, |N|$. The idea is to choose $\mathcal{U}$ such that both ultrapowers are as saturated as possible, and then use that saturated models are unique.

1. *Uniqueness of saturated models.* Two elementarily equivalent saturated structures of the same cardinality are isomorphic.
2. *Good ultrafilters.* If $\mathcal{U}$ is a countably incomplete λ^+-good ultrafilter over λ, then $\prod_{\mathcal{U}} \mathcal{M}$ and $\prod_{\mathcal{U}} \mathcal{N}$ are λ^+-saturated.
3. *Existence.* Such an ultrafilter $\mathcal{U}$ exists (shown by $\mathrm{cof}(2^\lambda) > \lambda$).
4. *Counting.* A λ^+-saturated structure has cardinality at least λ^+. On the other hand, $|\prod_{\mathcal{U}} \mathcal{M}| \leq |M|^\lambda \leq \lambda^\lambda = 2^\lambda = \lambda^+$, and similarly for $\mathcal{N}$. So, both ultrapowers are saturated structures of cardinality λ^+. They are elementarily equivalent since $\prod_{\mathcal{U}} \mathcal{M} \equiv \mathcal{M} \equiv \mathcal{N} \equiv \prod_{\mathcal{U}} \mathcal{N}$. By (1), they are isomorphic.

GCH is used only in (4).

The report is organized accordingly. We begin with filters and ultrafilters, and count the number of ultrafilters over a set. We then define ultraproducts and prove Łoś' theorem. Its applications include a proof of the Compactness Theorem and a description of ultraproducts of ultraproducts via product ultrafilters. Next, we recall saturated and universal models. Then, we study countably incomplete ultrafilters, whose ultraproducts are always ω_1-saturated, and the Rudin-Keisler ordering, which compares ultrafilters via the ultrapowers they give rise to. Finally, we introduce regular and good ultrafilters, prove that good ultrafilters yield saturated ultraproducts, show that they exist under GCH, and conclude with the proof of the Keisler-Shelah theorem.

We assume familiarity with first-order logic (structures, formulae, elementary embeddings, types) and basic cardinal arithmetic. We work in ZFC, and GCH is assumed only where explicitly mentioned.

2 Ultrafilters

A filter characterizes "large" subsets of a set. Throughout this section, let X denote a non-empty set.

Definition 2.1 (Filter)**.** $\mathcal{F} \subseteq \mathcal{P}(X)$ is a filter over X if the following hold.

(i) $\emptyset \notin \mathcal{F}$ and $X \in \mathcal{F}$.

(ii) $A, B \in \mathcal{F} \Rightarrow A \cap B \in \mathcal{F}$.

(iii) $A \in \mathcal{F}$ and $A \subseteq B$, then $B \in \mathcal{F}$.

Example. Let $a \in X$. Then $\mathcal{F} = \{A \in \mathcal{P}(X) : a \in A\}$ is a filter. It is called the *principal filter* at a.

Not all filters are so trivial, as we will see later. Some familiar collections of "large" subsets are also filters.

Example. Let X be infinite and $(Y, \mathcal{M}, \mu)$ be a complete measure space with $\mu(Y) > 0$. Then,

1. $\mathcal{F} = \{A \in \mathcal{P}(X) : |X \setminus A| < \omega\}$ is the *cofinite filter* over X.
2. $\mathcal{F} = \{A \in \mathcal{M} : \mu(Y \setminus A) = 0\}$ is the *conull filter* over Y.

Definition 2.2 (Ultrafilter)**.** A filter $\mathcal{F}$ over X is an ultrafilter if $(\forall A \in \mathcal{P}(X))(A \in \mathcal{F}$ or $X \setminus A \in \mathcal{F})$.

Remark. Note that we can never have $A \in \mathcal{U}$ as well as $X \setminus A \in \mathcal{U}$ for an ultrafilter $\mathcal{U}$ over X. Since this would imply $\emptyset = A \cap (X \setminus A) \in \mathcal{U}$. So, we may as well replace the "or" with an "exclusive or" above.

Proposition 2.3. Let $\mathcal{F}$ be a filter and $X \setminus A \notin \mathcal{F}$. Then $\mathcal{F}' = \{C : (\exists B \in \mathcal{F})(A \cap B \subseteq C)\}$ is a filter. Furthermore, $\mathcal{F}'$ extends $\mathcal{F}$ and $A \in \mathcal{F}$.

Proof. We verify each of the requirements (i)-(iii) from **Definition 2.1** for $\mathcal{F}'$.

(i) Suppose $\emptyset \in \mathcal{F}'$. Then, $\exists B \in \mathcal{F}$ such that $A \cap B = \emptyset$. But then $B \subseteq X \setminus A$, which implies $X \setminus A \in \mathcal{F}$. This contradicts our hypothesis, so $\emptyset \notin \mathcal{F}'$. Furthermore, $X \in \mathcal{F}$ and $A \cap X \subseteq X$. So, $X \in \mathcal{F}'$.

(ii) Suppose $C_1, C_2 \in \mathcal{F}'$. Choose $B_1, B_2 \in \mathcal{F}$ such that $A \cap B_1 \subseteq C_1$ and $A \cap B_2 \subseteq C_2$. But then, $B_1 \cap B_2 \in \mathcal{F}$ and $A \cap (B_1 \cap B_2) \subseteq C_1 \cap C_2$. So, $C_1 \cap C_2 \in \mathcal{F}'$.

(iii) Suppose $C \in \mathcal{F}'$ and $C \subseteq D$. Choose $B \in \mathcal{F}$ such that $A \cap B \subseteq C$. Then, $A \cap B \subseteq D$. Thus, $D \in \mathcal{F}'$.

Clearly, $(\forall B \in \mathcal{F})(A \cap B \subseteq B)$ and thus, $\mathcal{F} \subseteq \mathcal{F}'$. Furthermore, $X \in \mathcal{F}$ and $A \cap X \subseteq A$. So, $A \in \mathcal{F}'$. □

Proposition 2.4. A filter $\mathcal{U}$ over X is an ultrafilter $\Leftrightarrow$ $\mathcal{U}$ is a maximal filter over X.

Proof. ($\Rightarrow$) Suppose $\mathcal{U}$ is an ultrafilter over X that is not maximal. Fix another filter $\mathcal{U}_1 \supsetneq \mathcal{U}$ and $A \in \mathcal{U}_1 \setminus \mathcal{U}$. Since $A \notin \mathcal{U}$ and $\mathcal{U}$ is an ultrafilter, $X \setminus A \in \mathcal{U}$. But then, both $A, X \setminus A \in \mathcal{U}_1$. Contradiction.

($\Leftarrow$) Suppose $\mathcal{U}$ is a maximal filter over X and $A \in \mathcal{P}(X)$. Suppose both A and $X \setminus A$ are not in $\mathcal{U}$. Then by **Proposition 2.3**, we can extend $\mathcal{U}$ to $\mathcal{U}'$, where $A \in \mathcal{U}' \setminus \mathcal{U}$. Contradicts the fact that $\mathcal{U}$ was maximal. □

By Zorn's Lemma and **Proposition 2.4**, every filter can be extended to an ultrafilter. In particular, every infinite set has a non-principal ultrafilter over it (by extending the cofinite filter). How many ultrafilters are there over a set? We shall answer this question in the next section.

3 The number of ultrafilters

One may think of ultrafilters as the subsets of full measure on the measure space $(X, \mathcal{P}(X), \mu)$ where μ is a $\{0,1\}$-valued measure. So, if X is finite, this reduces to a some single point in X having full measure. In particular, we have the following.

Proposition 3.1. $|X| < \omega \Rightarrow$ all ultrafilters over X are principal.

Proof. $\mathcal{U}$ be an ultrafilter over X. Suppose $(\forall x \in X)(\exists A_x \in \mathcal{U})(x \notin A_x)$. Then $\bigcap_{x\in X} A_x = \emptyset \in \mathcal{U}$ since it is a finite intersection. But $\emptyset \notin \mathcal{U}$. Contradiction. So we must have that $(\exists x \in X)(\forall A \in \mathcal{U})(x \in A)$.

Suppose there are two such elements, $x, y \in X$. Consider $\{x\}, \{y\} \in \mathcal{P}(X)$. Clearly, both of them cannot be in $\mathcal{U}$. If neither of them are in $\mathcal{U}$, then $\{x\}^c \cap \{y\}^c$ is in $\mathcal{U}$, but it contains neither x or y. Contradiction. So exactly one of $\{x\}$ and $\{y\}$ must be in $\mathcal{U}$. But this contradicts the existence of two such elements.

So $(\exists! x \in X)(\forall A \in \mathcal{U})(x \in A)$ and furthermore, $\{x\} \in \mathcal{U}$. So, $\mathcal{U}$ is principal at $x \in X$. □

Corollary 3.2. $|X| = n < \omega \Rightarrow$ there are n ultrafilters over X.

The case for ultrafilters over infinite sets is quite different. One immediately sees that there are atmost 2^{2^κ} ultrafilters over X if $|X| = \kappa$, for some infinite cardinal κ. This is indeed achieved.

Definition 3.3 (Finite Intersection Property)**.** A collection $\mathcal{E}$ of subsets of X is said to have the finite intersection property (FIP) if any finite subcollection of $\mathcal{E}$ has non-empty intersection.

Lemma 3.4. $\mathcal{E}$ has FIP $\Rightarrow \mathcal{E}$ extends to a filter over X, hence to an ultrafilter over X.

Proof. Define $\mathcal{F} = \{B \subseteq Y : (\exists n < \omega)(n \neq 0)(\exists E_1, \ldots, E_n \in \mathcal{E})(E_1 \cap \ldots \cap E_n \subseteq B)\}$. Clearly $\mathcal{E} \subseteq \mathcal{F}$, and $\mathcal{F}$ is closed under finite intersection and supersets. Also, $\emptyset \notin \mathcal{F}$ as this would contradict the FIP of $\mathcal{E}$. □

Theorem 3.5. $|X| = \kappa \Rightarrow$ there are 2^{2^κ} ultrafilters on X.

Proof. Let $\mathcal{F} = [X]^{<\omega}$. Then $|\mathcal{F} \times [\mathcal{F}]^{<\omega}| = \kappa$. We will define 2^{2^κ} ultrafilters over $\mathcal{F} \times [\mathcal{F}]^{<\omega}$. Let $s \in \mathcal{P}(X)$.

$$b_s := \{(F, G) \in \mathcal{F} \times [\mathcal{F}]^{<\omega} : s \cap F \in G\}$$

Define $b_s^c := (\mathcal{F} \times [\mathcal{F}]^{<\omega}) \setminus b_s$. Now for each of the 2^{2^κ} subsets $\mathcal{S} \subseteq \mathcal{P}(X)$, define

$$\mathfrak{B}_\mathcal{S} = \{b_s : s \in \mathcal{S}\} \cup \{b_s^c : s \notin \mathcal{S}\}$$

Claim: $\mathfrak{B}_\mathcal{S}$ has FIP.

Proof of claim. Let $b_{s_1}, \ldots, b_{s_n}, b^c_{s_{n+1}}, \ldots, b^c_{s_m}$ be distinct members of $\mathfrak{B}_\mathcal{S}$. Each s_i are distinct subsets of X. For each $i < j$, choose $x_{ij} \in X$ that belongs to exactly one of s_i, s_j. Then $F = \{x_{ij} : 1 \leq i < j \leq m\}$ is an element of $\mathcal{F}$. Define $G = \{s_1 \cap F, \ldots, s_n \cap F\} \in [\mathcal{F}]^{<\omega}$. Then clearly $(F, G) \in \bigcap_{i\leq n} b_{s_i} \cap \bigcap_{n<i\leq m} b^c_{s_i}$. □

Therefore by **Lemma 3.4**, each $\mathfrak{B}_\mathcal{S}$ extends to an ultrafilter over $\mathcal{F} \times [\mathcal{F}]^{<\omega}$. Suppose $\mathcal{S}, \mathcal{S}' \subseteq \mathcal{P}(X)$ are two distinct families. Then for $s \in \mathcal{S} \setminus \mathcal{S}'$, $\mathfrak{B}_\mathcal{S}$ contains b_s while $\mathfrak{B}_{\mathcal{S}'}$ contains b_s^c. Hence, the ultrafilters they extend to are distinct. Thus, we have 2^{2^κ} ultrafilters over κ. □

4 Łoś' theorem

Fix an ultrafilter $\mathcal{U}$ over I. Let $\{\mathcal{M}_i : i \in I\}$ be a collection of $\mathcal{L}$-structures. For $f, g \in \prod_I M_i$, define the relation $f \sim_{\mathcal{U}} g \Leftrightarrow \{i \in I : f(i) = g(i)\} \in \mathcal{U}$. Then, $\sim_{\mathcal{U}}$ is an equivalence relation on $\prod_I M_i$. Denote by $\prod_{\mathcal{U}} M_i$, the set of $\sim_{\mathcal{U}}$ equivalence classes.

Notation. We will use f to denote both the function $f \in \prod_I M_i$ as well as the $\sim_{\mathcal{U}}$-equivalence class of f.

Definition 4.1 (Ultraproduct)**.** The ultraproduct $\prod_{\mathcal{U}} \mathcal{M}_i$ is the unique $\mathcal{L}$-structure with domain $\prod_{\mathcal{U}} M_i$ with the following interpretations. Denote by N, the domain $\prod_{\mathcal{U}} M_i$.

(i) For every constant symbol c of $\mathcal{L}$, $c^{\mathcal{N}} = f$ where $f(i) = c^{\mathcal{M}_i}$ for all $i \in I$.

(ii) For every relation symbol R of $\mathcal{L}$, $R^{\mathcal{N}} = \{\overline{f} : \{i : \overline{f(i)} \in R^{\mathcal{M}_i}\} \in \mathcal{U}\}$.

(iii) For every n-ary function symbol F of $\mathcal{L}$, $F^{\mathcal{N}}(\overline{f}) = g$ where $\{i \in I : F^{\mathcal{M}_i}(\overline{f(i)}) = g(i)\} \in \mathcal{U}$.

Ultraproducts are like taking an average of the structures $\{\mathcal{M}_i : i \in I\}$. The truth value of an $\mathcal{L}$-formula φ is decided by looking at the truth value of φ on the individual structures $\mathcal{M}_i$. The precise statement is the following theorem due to Łoś. It is also sometimes called the *Fundamental Theorem of Ultraproducts*.

Theorem 4.2 (Łoś)**.** Let $\varphi(x_1, \ldots, x_n)$ be an $\mathcal{L}$-formula and $f_1, \ldots, f_n \in \prod_{\mathcal{U}} M_i$. Then

$$\prod_{\mathcal{U}} \mathcal{M}_i \models \varphi(f_1, \ldots, f_n) \Leftrightarrow \{i : \mathcal{M}_i \models \varphi(f_1(i), \ldots, f_n(i))\} \in \mathcal{U}$$

Proof. The proof is via induction on the length of φ. We skip the tedious details. □

Corollary 4.3. T be an $\mathcal{L}$-theory. If $\mathcal{M}_i \models T$ for all $i \in I$ then $\prod_{\mathcal{U}} \mathcal{M}_i \models T$.

Definition 4.4 (Ultrapower)**.** The ultraproduct $\prod_{\mathcal{U}} \mathcal{M}$ is called the $\mathcal{U}$-ultrapower of $\mathcal{M}$.

Let $\mathcal{M}$ be an $\mathcal{L}$-structure. Let $a \in M$. By $f_a \in \prod_{i \in I} M$, denote the constant function $f_a(i) = a$ for all $i \in I$ The *diagonal embedding* is the map $d : M \to \prod_{\mathcal{U}} M$ defined as $d(a) = f_a$ where f_a denotes the $\sim_{\mathcal{U}}$-equivalence class of the same function. The diagonal embedding is an elementary embedding.

Corollary 4.5. $d : \mathcal{M} \prec \prod_{\mathcal{U}} \mathcal{M}$ and if $\mathcal{M}$ is finite, then $d : \mathcal{M} \cong \prod_{\mathcal{U}} \mathcal{M}$.

Proof. Let $\varphi(x_1, \ldots, x_n)$ be an $\mathcal{L}$-formula and $a_1, \ldots, a_n \in M$. Then $\mathcal{M} \models \varphi(a_1, \ldots, a_n)$ implies that for all $i \in I$, $\mathcal{M} \models \varphi(f_{a_1}(i), \ldots, f_{a_n}(i))$. Since $I \in \mathcal{U}$, by Łoś' theorem $\prod_{\mathcal{U}} \mathcal{M} \models \varphi(f_{a_1}, \ldots, f_{a_n})$. Conversely, suppose $\prod_{\mathcal{U}} \mathcal{M} \models \varphi(f_{a_1}, \ldots, f_{a_n})$. Now, either $\mathcal{M} \models \varphi(a_1, \ldots, a_n)$ or $\mathcal{M} \not\models \varphi(a_1, \ldots, a_n)$. In the second case, $\{i \in I : \mathcal{M} \models \neg\varphi(f_{a_1}(i), \ldots, f_{a_n}(i))\} = I \in \mathcal{U}$. By Łoś' theorem, $\prod_{\mathcal{U}} \mathcal{M} \models \neg\varphi(f_{a_1}, \ldots, f_{a_n})$. Contradiction.

Since d is elementary, it is injective. Suppose $M = \{a_1, \ldots, a_n\}$. Let φ be the sentence $\forall x \bigvee_{i=1}^{n} (x = a_i)$. Since $\mathcal{M} \models \varphi$, we have $\prod_{\mathcal{U}} \mathcal{M} \models \forall x \bigvee_{i=1}^{n} (x = d(a_i))$. This shows that d is surjective, and thus an isomorphism. □

5 Applications of Łoś' theorem

An important consequence of Łoś' theorem, is that it gives a "constructive" proof of the Compactness Theorem. The required model is directly obtained via ultraproducts.

Theorem 5.1 (Compactness)**.** T be an $\mathcal{L}$-theory. T is satisfiable $\Leftrightarrow$ T is finitely satisfiable.

Proof. ($\Rightarrow$) Trivial. ($\Leftarrow$) Let $I = [T]^{<\omega}$ and for each $i \in I$, let $\mathcal{M}_i \models i$. For each $i \in I$, define $X_i = \{j \in I : i \subseteq j\}$. Clearly $i \in X_i$ and $X_{i\cup j} = X_i \cap X_j$. Define $\mathcal{F} = \{X \subseteq I : X_i \subseteq X \text{ for some } i \in I\}$.

We check that $\mathcal{F}$ is a filter:

(i) For each $X \in \mathcal{F}$, there is some $i \in I$ such that $X_i \subseteq X$ and $i \in X_i$. So $X \neq \emptyset$. Thus, $\emptyset \notin \mathcal{F}$. Also, I contains all X_i's, so $I \in \mathcal{F}$.

(ii) Suppose $X, Y \in \mathcal{F}$. Fix $i, j \in I$ such that $X_i \subseteq X$ and $X_j \subseteq Y$. Then $X_{i\cup j} = X_i \cap X_j \subseteq X \cap Y$. So, $X \cap Y \in \mathcal{F}$.

(iii) Suppose $X \in \mathcal{F}$ and $X \subseteq Y$. Let $i \in I$ such that $X_i \subseteq X$. Then $X_i \subseteq Y$, which implies $Y \in \mathcal{F}$.

$\mathcal{F}$ extends to an ultrafilter $\mathcal{U}$ over I. We claim that $\prod_{\mathcal{U}} \mathcal{M}_i \models T$. Let $\varphi \in T$ and note that $X_{\{\varphi\}} \in \mathcal{U}$. Now, $i \in X_{\{\varphi\}} \Rightarrow \varphi \subseteq i$. Since $\mathcal{M}_i \models i$, we have $\mathcal{M}_i \models \varphi$. Thus,

$$X_{\{\varphi\}} \subseteq \{i \in I : \mathcal{M}_i \models \varphi\} \text{ and } X_{\{\varphi\}} \in \mathcal{U} \quad \Rightarrow \quad \{i \in I : \mathcal{M}_i \models \varphi\} \in \mathcal{U}$$

By Łoś' theorem, $\prod_{\mathcal{U}} \mathcal{M}_i \models \varphi$ and thus, $\prod_{\mathcal{U}} \mathcal{M}_i \models T$. □

The above proof of the Compactness Theorem is "algebraic" in the sense that it is "logic-free" or free of syntactic notions. We shall see more results of similar flavor that translate algebraic properties to syntactic ones and vice-versa in later sections.

Certain relationships between structures are preserved under taking ultraproducts. One such example is expansions, which we prove below. The key ingredient again, is Łoś' theorem.

Proposition 5.2 (Expansion Property)**.** Let $\mathcal{L}_1 \subseteq \mathcal{L}_2$. $\mathcal{M}_i$ be an $\mathcal{L}_1$-structure with an $\mathcal{L}_2$-expansion $\mathcal{N}_i$ for each $i \in I$. Then for every ultrafilter $\mathcal{U}$ over I, $\prod_{\mathcal{U}} \mathcal{N}_i$ is an $\mathcal{L}_2$-expansion of $\prod_{\mathcal{U}} \mathcal{M}_i$.

Proof. Let $\mathcal{M}^* = \prod_{\mathcal{U}} \mathcal{M}_i$ and $\mathcal{N}^* = \prod_{\mathcal{U}} \mathcal{N}_i$. Since each $\mathcal{N}_i$ is an expansion of $\mathcal{M}_i$, they have the same universe, M_i. Consequently, $\mathcal{N}^*$ has the same universe as $\mathcal{M}^*$; the $\sim_{\mathcal{U}}$-equivalence classes of $\prod_I M_i$.

Let R be an $\mathcal{L}_1$ unary relation symbol. Then $f \in R^{\mathcal{M}^*} \Leftrightarrow \{i \in I : f(i) \in R^{\mathcal{M}_i}\} \in \mathcal{U}$. Since $\mathcal{N}_i$ and $\mathcal{M}_i$ agree on the interpretations of $\mathcal{L}_1$ symbols, $R^{\mathcal{M}_i} = R^{\mathcal{N}_i}$. Also, $f \in R^{\mathcal{N}^*} \Leftrightarrow \{i \in I : f(i) \in R^{\mathcal{N}_i}\} \in \mathcal{U}$. This shows $R^{\mathcal{M}^*} = R^{\mathcal{N}^*}$. The verification for constant, function symbols is similar, and we skip the tedious details. □

Let $\mathcal{M}, \mathcal{N}$ be two $\mathcal{L}$-structures. Given an ultrafilter $\mathcal{U}$ over I and a map $h : M \to N$, define

$$\prod_{\mathcal{U}} h : \prod_{\mathcal{U}} M \to \prod_{\mathcal{U}} N \quad \text{as the map} \quad f \mapsto h \circ f$$

where $f \in \prod_{\mathcal{U}} M$ and $h \circ f \in \prod_{\mathcal{U}} N$ are the respective $\mathcal{U}$-equivalence classes. Then as a consequence of Łoś' Theorem, certain properties of $h : M \to N$ are carried over to the ultrapowers.

Proposition 5.3. The map $h \mapsto \prod_{\mathcal{U}} h$ is a functor on the category of homomorphisms between $\mathcal{L}$-structures. We also have:

(i) h is surjective $\Rightarrow \prod_{\mathcal{U}} h$ is surjective.

(ii) h is an isomorphism $\Rightarrow \prod_{\mathcal{U}} h$ is an isomorphism.

(iii) h is an elementary embedding $\Rightarrow \prod_{\mathcal{U}} h$ is an elementary embedding.

Proof. **well-definedness.** Suppose $f \sim_{\mathcal{U}} f'$. Then $\{i \in I : f(i) = f'(i)\} \in \mathcal{U}$. Then $h \circ f \sim_{\mathcal{U}} h \circ f'$ is given by $\{i \in I : h(f(i)) = h(f'(i))\} \supseteq \{i \in I : f(i) = f'(i)\}$. So, $f \mapsto h \circ f$ is well-defined on $\mathcal{U}$-classes.

functoriality. Clearly $\prod_{\mathcal{U}} \mathrm{id}_M = \mathrm{id}_{\prod_{\mathcal{U}} M}$. Suppose $h : M \to N$ and $h' : N \to K$ are homomorphisms. $(\prod_{\mathcal{U}} h')(\prod_{\mathcal{U}} h)(f) = (\prod_{\mathcal{U}} h')(h \circ f) = h' \circ h \circ f = (\prod_{\mathcal{U}} h' \circ h)(f)$.

Let $\mathcal{M}^* = \prod_{\mathcal{U}} \mathcal{M}$ and $\mathcal{N}^* = \prod_{\mathcal{U}} \mathcal{N}$.

$\prod_{\mathcal{U}} h$ **is a homomorphism.** We verify this only for constant and relation symbols.

1. Let c be a constant symbol. Then $c^{\mathcal{M}^*}$ is the constant function $c^{\mathcal{M}}$. Since h is a homomorphism, it's composition with the constant function, is the constant function $h(c^{\mathcal{M}}) = c^{\mathcal{N}}$ which is $c^{\mathcal{N}^*}$.
2. Let R be a relation symbol. Suppose $f \in R^{\mathcal{M}^*}$. Since $h : M \to N$ is a homomorphism, $f(i) \in R^{\mathcal{M}}$ implies $f(i) \in R^{\mathcal{N}}$. So, $\{i \in I : f(i) \subseteq R^{\mathcal{M}}\} \subseteq \{i \in I : f(i) \in R^{\mathcal{N}}\}$ which implies $f \in R^{\mathcal{N}^*}$

We now check the three claims.

(i) Suppose h is surjective and $g \in \prod_{\mathcal{U}} N$. For each $i \in I$, let $f(i) \in M$ be such that $h(f(i)) = g(i)$. Then $\prod_{\mathcal{U}} h(f) = g$ which shows $\prod_{\mathcal{U}} h$ is surjective.

(ii) Suppose h is an isomorphism. Then by the functoriality shown above, one can check $\prod_{\mathcal{U}}(h^{-1})$ is the inverse $\prod_{\mathcal{U}} h$. This shows that $\prod_{\mathcal{U}} h$ is also an isomorphism.

(iii) Suppose h is an elementary embedding. By Łoś' theorem

$$\prod_{\mathcal{U}} \mathcal{M} \models \varphi(f) \Leftrightarrow \{i : \mathcal{M} \models \varphi(f(i))\} \in \mathcal{U} \quad \text{and} \quad \prod_{\mathcal{U}} \mathcal{N} \models \varphi(h \circ f) \Leftrightarrow \{i : \mathcal{N} \models \varphi(h(f(j)))\} \in \mathcal{U}$$

Since h is elementary, the above two sets coincide. Thus $\prod_{\mathcal{U}} \mathcal{M} \models \varphi(f) \Leftrightarrow \prod_{\mathcal{U}} \mathcal{N} \models \varphi(h(f))$. This shows that $\prod_{\mathcal{U}} h$ is an elementary embedding.

□

Next we consider ultraproduct of ultraproducts. This turns out to be isomorphic to a single ultraproduct. We have the following definition.

Definition 5.4 (Product ultrafilter)**.** $\mathcal{U}, \mathcal{V}$ be ultrafilters over I, J respectively. Then

$$\mathcal{U} \times \mathcal{V} := \{Y \subseteq I \times J : \{j \in J : Y_j \in \mathcal{U}\} \in \mathcal{V}\}$$

where $Y_j = \{i \in I : (i, j) \in Y\}$ is the vertical section of Y at j.

Proposition 5.5. $\mathcal{U} \times \mathcal{V}$ is an ultrafilter over $I \times J$.

Proof. We verify each of the requirements.

(i) Clearly $(\emptyset)_j = \emptyset \notin \mathcal{U}$ for all $j \in J$. So, $\{j \in J : (\emptyset)_j \in \mathcal{U}\} = \emptyset \notin \mathcal{V}$. This implies $\emptyset \notin \mathcal{U} \times \mathcal{V}$. Also, $(I \times J)_j = I \in \mathcal{U}$ for all $j \in J$. So, $\{j \in J : (I \times J)_j \in \mathcal{U}\} = J \in \mathcal{V}$. This implies $I \times J \in \mathcal{U} \times \mathcal{V}$.

(ii) Let $Y, Z \in \mathcal{U} \times \mathcal{V}$. Suppose $Y_j, Z_j \in \mathcal{U}$. Then $(Y \cap Z)_j = Y_j \cap Z_j \in \mathcal{U}$.

Now, note that $\{j \in J : Y_j \in \mathcal{U}\} \cap \{j \in J : Z_j \in \mathcal{V}\} \subseteq \{j \in J : (Y \cap Z)_j \in \mathcal{V}\}$. Since the LHS is the intersection of two sets in $\mathcal{V}$, the RHS must be in $\mathcal{V}$. This implies $Y, Z \in \mathcal{U} \times \mathcal{V}$.

(iii) Suppose $Y \in \mathcal{U} \times \mathcal{V}$ and $Y \subseteq Z$. Then $Y_j \subseteq Z_j$ and thus, $Y_j \in \mathcal{V} \Rightarrow Z_j \in \mathcal{V}$. Since $\{j \in J : Y_j \in \mathcal{V}\} \in \mathcal{U}$, it's superset $\{j \in J : Z_j \in \mathcal{V}\}$ must also be in $\mathcal{V}$. Thus, $Z \in \mathcal{U} \times \mathcal{V}$.

(iv) Note that $((I \times J) \setminus Y)_j = I \setminus Y_j$ for $Y \subseteq I \times J$. Since $\mathcal{U}$ is an ultrafilter, for all $j \in J$, either $Y_j \in \mathcal{U}$ or $I \setminus Y_j \in \mathcal{U}$, i.e. $((I \times J) \setminus Y)_j \in \mathcal{U}$. So we have

$$\{j \in J : Y_j \in \mathcal{U}\} = J \setminus \{j \in J : ((I \times J) \setminus Y)_j \in \mathcal{U}\}$$

Since $\mathcal{V}$ is an ultrafilter, one of the above must be in $\mathcal{V}$. If the LHS is in $\mathcal{V}$, then $Y \in \mathcal{U} \times \mathcal{V}$ and if the RHS is in $\mathcal{V}$, then $(I \times J) \setminus Y \in \mathcal{V}$. So, $\mathcal{U} \times \mathcal{V}$ is an ultrafilter.

□

Proposition 5.6. $\mathcal{U}, \mathcal{V}$ be ultrafilters over I, J respectively. $\mathcal{M}$ be any $\mathcal{L}$-structure. Then:

(i) $\prod_{\mathcal{U}\times\mathcal{V}} \mathcal{M} \cong \prod_{\mathcal{V}} (\prod_{\mathcal{U}} \mathcal{M})$.

(ii) Both $\prod_{\mathcal{U}} \mathcal{M}$ and $\prod_{\mathcal{V}} \mathcal{M}$ are elementarily embeddable in $\prod_{\mathcal{U}\times\mathcal{V}} \mathcal{M}$.

Proof. (i) Define the map $\Phi : \prod_{\mathcal{U}\times\mathcal{V}} M \to \prod_{\mathcal{V}}(\prod_{\mathcal{U}} M)$ as follows. Suppose $f : I \times J \to M$. Let $\hat{f} : J \to \prod_{\mathcal{U}} M$ be the function $\hat{f}(j) = (i \mapsto f(i,j))$. One can check that the map $f \mapsto \hat{f}$ is well-defined on $\mathcal{U} \times \mathcal{V}$-classes and $\Phi(f) = \hat{f}$ is an isomorphism.

(ii) By (i), it sufficient to elementarily embed $\prod_{\mathcal{U}} \mathcal{M}$ and $\prod_{\mathcal{V}} \mathcal{M}$ into $\prod_{\mathcal{V}}(\prod_{\mathcal{U}} \mathcal{M})$ and then compose the elementary embeddings with Φ^{-1}. Let $\mathcal{N} = \prod_{\mathcal{U}} \mathcal{M}$. Then by **Corollary 4.5**, $d : \mathcal{N} \to \prod_{\mathcal{V}} \mathcal{N}$ is an elementary embedding. This handles the case for $\prod_{\mathcal{U}} \mathcal{M}$.

For $\prod_{\mathcal{V}} \mathcal{M}$, let $d : \mathcal{M} \to \prod_{\mathcal{U}} \mathcal{M}$ be the diagonal embedding. By **Proposition 5.3**, $\prod_{\mathcal{V}} d : \prod_{\mathcal{V}} \mathcal{M} \to \prod_{\mathcal{V}}(\prod_{\mathcal{U}} \mathcal{M})$ is an elementary embedding. This suffices. □

Remark. One can see that both $\mathcal{U}$ and $\mathcal{V}$ are simpler than $\mathcal{U} \times \mathcal{V}$ in the sense that all the ultrapowers with respect to the former, elementarily embeds into ultrapowers of the latter. This hints at an ordering of ultrafilters, where $\mathcal{V} \leq \mathcal{U}$ whenever $\mathcal{V}$ is "simpler" than $\mathcal{U}$ in the above sense. We shall see this in **Section 8**.

6 Saturated and universal models

In this section we recall some definitions and results from model theory.

Let $\mathcal{M}$ be an $\mathcal{L}$-structure and $A \subseteq M$. An n-type p is a set of $\mathcal{L}_A$-formulae in free variables $x_1, \ldots, x_n$ such that $p \cup \mathrm{Th}_A(\mathcal{M})$ is satisfiable. An n-type is *complete* if for all $\mathcal{L}_A$-formulae, either $\phi \in p$ or $\neg\phi \in p$. $S_n^{\mathcal{M}}(A)$ is the set of all complete n-types. If the underlying structure is clear from the context, then we will use $S_n(A)$ to denote the same.

Intuitively, saturation is a measure of richness of models, in terms of the types that it realizes. More saturated models, realizes more complex types. In the following discussion, κ denotes an infinite cardinal. $\mathcal{L}$ denotes the underlying language and $\mathcal{M}, \mathcal{N}$ denote $\mathcal{L}$-structures with domains M, N respectively.

Proposition 6.1. $\mathcal{M}$ be an $\mathcal{L}$-structure and $A \subseteq M$. Every $p \in S_1(A)$ is realized in some $\mathcal{N} \succeq \mathcal{M}$

Proof. Let $\mathcal{L}' = \mathcal{L}_M \cup \{c\}$. Then $p' = p(c) \cup \mathrm{Th}_M(\mathcal{M})$ is a set of $\mathcal{L}'$-sentences. Clearly, p' is finitely-satisfiable. By Compacntess Theorem, p' has a model $\mathcal{N}'$. Let $\mathcal{N}$ be the $\mathcal{L}$-reduct of $\mathcal{N}'$. □

Example. $\mathcal{M} = (\omega, <)$ and $p(x) = \{n < x : n < \omega\}$. Clearly, p is finitely-satisfiable in $\mathcal{M}$ but not satisfiable.

The above example illustrates that sometimes one cannot avoid elementary extensions while trying to realize types. A saturated structure is one where this is avoidable and we can realize any type in the structure itself (provided the set of parameters is small).

Definition 6.2 (Saturation)**.** $\mathcal{M}$ is κ-saturated if $(\forall A \in [M]^{<\kappa})$(every 1-type over A is realized in $\mathcal{M}$).

Example. $\mathcal{M} = (\mathbb{Q}, <)$ is ω-saturated.

Proposition 6.3. Let $\mathcal{M}$ be κ-saturated. Then the following are true.

(i) $\mathcal{M}_A$ is κ-saturated for all $A \subseteq M$ with $|A| < \kappa$.

(ii) $(\forall A \in [M]^{<\kappa})$(every n-type over A is realized in $\mathcal{M}$).

(iii) $\mathcal{M}$ is κ-saturated $\Rightarrow |M| \geq \kappa$.

Proof. We only prove (iii). Consider the 1-type $p(x) = \{x \neq a : a \in M\}$ over M. If $|M| < \kappa$, by κ-saturation, p is realized in $\mathcal{M}$. That is, $\exists b \in M$ such that $\mathcal{M} \models p(b)$. But then, $(\forall a \in M)(a \neq b)$. Contradiction. □

We now recall some results from model theory (without proof).

Theorem 6.4. $\mathcal{M}, \mathcal{N}$ are saturated with $|M| = |N|$ and $M \equiv N$. Then $\mathcal{M} \cong \mathcal{N}$.

Universality is a weakening of saturation.

Definition 6.5 (Universality)**.** $\mathcal{M}$ is κ-universal if $\mathcal{M} \equiv \mathcal{N}$ and $|N| < \kappa \Rightarrow \mathcal{N} \prec \mathcal{M}$.

Proposition 6.6. $\mathcal{M}$ is κ-saturated $\Rightarrow \mathcal{M}$ is κ^+-universal.

Proposition 6.7. Suppose $|\mathcal{L}| \leq \kappa$. Then an $\mathcal{L}$-structure $\mathcal{M}$ is κ-saturated $\Leftrightarrow \forall A \in [M]^{<\kappa}, \mathcal{M}_A$ is κ^+-universal.

Proposition 6.8. Suppose $|\mathcal{L}| \leq \kappa$. For an $\mathcal{L}$-structure $\mathcal{M}$, the following are equivalent:

(i) $\mathcal{M}$ is κ^+-universal.

(ii) For any set p of $\mathcal{L}(\kappa)$-sentences, if $\mathrm{Th}(\mathcal{M}) \cup p$ is satisfiable, then p is satisfiable in $(\mathcal{M}, a)$ for some $a \in M^\kappa$.

7 Uniform and countably incomplete ultrafilters

Definition 7.1 (Uniform ultrafilter)**.** An ultrafilter $\mathcal{U}$ over I is uniform if $(\forall X \in \mathcal{U})(|X| = |I|)$.

If I is a singleton, then the trivial filter $\mathcal{U} = \{I\}$ is a uniform ultrafilter. But for finite I with cardinality > 1, all ultrafilters are principal by **Proposition 3.1**. All of them contain some singleton set, and thus are not uniform. Now if I is infinite and $\mathcal{U}$ is an ultrafilter over I, it must contain all cofinite sets (since it must contain a set or it's complement). We show that this is sufficient.

Proposition 7.2. An ultrafilter $\mathcal{U}$ over I (infinite) is uniform $\Leftrightarrow F := \{X \subseteq I : |I \setminus X| < |I|\} \subseteq \mathcal{U}$.

Proof. $(\Rightarrow)$ is clear from the above. For $(\Rightarrow)$, suppose $X \in \mathcal{U}$ and $|X| < |I|$. Let $Y = I \setminus X$. Then $Y \in F$, which implies $Y \in \mathcal{U}$. But then $\emptyset = X \cap Y \in \mathcal{U}$. Contradiction. □

Now since F defined above (subsets with small complement) is a filter, it can be extended to an ultrafilter. So uniform ultrafilter over infinite sets exists. The next result shows that for ultraproducts, it is enough to only consider uniform ultrafilters.

Proposition 7.3. $\prod_{\mathcal{U}} \mathcal{M}_i$ be any ultraproduct. Then $\prod_{\mathcal{U}} \mathcal{M}_i \cong \prod_{\mathcal{V}} \mathcal{M}_i$ where $\mathcal{V}$ is uniform.

Proof. Suppose $\mathcal{U}$ is a non-uniform ultrafilter over I. Let $J \in \mathcal{U}$ have minimum cardinality. Then $\mathcal{V} = \mathcal{U} \cap \mathcal{P}(J)$ is a uniform ultrafilter over J. We check the four requirements from **Definition 2.1 & 2.2**.

(i) Clearly $\emptyset \notin \mathcal{V}$ and $J \in \mathcal{V}$.

(ii) Suppose $Y, Z \in \mathcal{V} = \mathcal{U} \cap \mathcal{P}(J)$. Then $Y \cap Z \in \mathcal{U}$ and $Y \cap Z \in \mathcal{P}(J)$. So, $Y \cap Z \in \mathcal{V}$.

(iii) Suppose $Y \in \mathcal{V}$ and $Y \subseteq Z \subseteq J$. Since $\mathcal{U}$ is a filter, and $Y \in \mathcal{U}$, we have $Z \in \mathcal{U}$. Also $Z \in \mathcal{P}(J)$, by definition. So $Z \in \mathcal{V}$.

(iv) Suppose $Y \subseteq J$. Either $Y \in \mathcal{U}$ or $I \setminus Y \in \mathcal{U}$. In the first case $Y \in \mathcal{V}$ and in the second case $J \setminus Y = J \cap (I \setminus Y) \in \mathcal{V}$.

If $X \in \mathcal{V}$, then $|X| = |J|$. This is because $X \subseteq J$ and J had the minimum cardinality among sets in $\mathcal{U}$. □

Definition 7.4 (Countably incomplete ultrafilters)**.** An ultrafilter $\mathcal{U}$ is countably incomplete if it has a countable subset $\mathcal{E} \subseteq \mathcal{U}$ such that $\bigcap \mathcal{E} = \emptyset$.

Example. Any non-principal ultrafilter over ω is countably incomplete. This is because $\omega \setminus \{n\} \in \mathcal{U}$ for all $n < \omega$. Take $\mathcal{E} = \{\omega \setminus \{n\} : n < \omega\}$. Clearly $\bigcap \mathcal{E} = \emptyset$.

We can always assume that the witnessing set $\mathcal{E} \subseteq \mathcal{U}$ has decreasing sets. That is, $\mathcal{E} = \{I_n : n < \omega\}$ such that $I_{n+1} \subseteq I_n$ (by taking finite intersections). We can also further assume that $I_0 = I$.

Proposition 7.5. $\mathcal{U}$ be a countably incomplete ultrafilter over I. $\mathcal{M}$ be an infinite $\mathcal{L}$-structure. Then $\prod_{\mathcal{U}} \mathcal{M}$ is (isomorphic to) a proper elementary extension of $\mathcal{M}$.

Proof. Let $\mathcal{E} = \{I_n : n < \omega\}$ witness that $\mathcal{U}$ is countably incomplete. Since M is infinite, fix pairwise distinct elements $\{a_n : n < \omega\}$ in M. Since I_n's are decreasing and $I_0 = I$, the set $\{n < \omega : i \in I_n\}$ is non-empty and bounded for all $i \in I$. Let $n(i) = \max\{n < \omega : i \in I_n\}$. Define $g(i) = a_{n(i)}$. Then $g \in \prod_{\mathcal{U}} M$.

Claim: (the equivalence class of) g is not in the range of the diagonal embedding $d : M \to \prod_{\mathcal{U}} M$.

Proof of claim. Suppose $g = d(a)$ for some $a \in M$. Then $\{i \in I : g(i) = a\} \in \mathcal{U}$. Then $a = a_m$ for some $m < \omega$. Then $\{i \in I : g(i) = a_m\} = \{i \in I : n(i) = m\} = I_m \setminus I_{m+1} \in \mathcal{U}$. But $I_{m+1} \in \mathcal{U}$. This implies $\emptyset = I_{m+1} \cap (I_m \setminus I_{m+1}) \in \mathcal{U}$. Contradiction. So, $g \neq d(a)$ for all $a \in M$. □

So d is not surjective and thus, $\prod_{\mathcal{U}} \mathcal{M}$ is a proper elementary extension of $\mathcal{M}$. □

We state some results about ultraproducts with respect to countably incomplete ultrafilters. The proofs of these are beyond the scope of this report. Throughout, let $\mathcal{U}$ be any countably incomplete ultrafilter over I and $\{\mathcal{M}_i : i \in I\}$ is a collection of $\mathcal{L}$-structures.

Theorem 7.6 (Frayne, Morel, and Scott)**.** $\prod_{\mathcal{U}} \mathcal{M}_i$ is either finite or of cardinality $\geq 2^{\omega}$.

Theorem 7.7 (Keisler, Shelah)**.** If $\prod_{\mathcal{U}} \mathcal{M}_i$ is infinite, then $|\prod_{\mathcal{U}} \mathcal{M}_i| = |\prod_{\mathcal{U}} \mathcal{M}_i|^{\omega}$.

Theorem 7.8 (Keisler)**.** $\prod_{\mathcal{U}} \mathcal{M}_i$ is ω_1-saturated, provided that the language $\mathcal{L}$ is countable.

Proof. Let $\mathcal{N} = \prod_{\mathcal{U}} \mathcal{M}_i$. Suppose $A \subseteq N$ is countable and $p \in S_1^{\mathcal{N}}(A)$.

Wlog, we can assume that $p = \{\varphi_n : n < \omega\}$ and $\mathcal{N} \models \forall x(\varphi_{n+1}(x) \Rightarrow \varphi_n(x))$. Let $\mathcal{E} = \{I_n : n < \omega\}$ witness that $\mathcal{U}$ is countably incomplete. Let $J_n = \{i \in I_n : \mathcal{M}_i \models \exists x \varphi_n(x)\}$. Since p is consistent, $J_n \in \mathcal{U}$. Also we can assume, $J_{n+1} \subseteq J_n$ and $\bigcap_{n<\omega} J_n = \emptyset$.

Now for each $i \in I$, let $n(i)$ be the maximum n satisfying $i \in J_n$. Let $b(i) \in M_i$ be the witness for $\mathcal{M}_i \models \exists x \varphi_n(x)$. Then $\mathcal{N} \models \varphi_n(b)$ for all $n < \omega$. Note that $i \in J_n \Rightarrow n \leq n(i)$. So $\mathcal{M}_i \models \varphi_{n(i)}(b(i))$, which implies $\mathcal{M}_i \models \varphi_n(b(i))$. Since $J_n \in \mathcal{U}$, by Łoś' theorem $\mathcal{N} \models \varphi_n(b)$. □

Remark. If $\mathcal{U}$ is not countably incomplete, then $\mathcal{U}$ is closed under countable intersections. Such ultrafilters are called *countably complete*. Every principal filter is countably complete. However, it is not provable from ZFC that non-principal countably complete ultrafilters exist.

8 Rudin-Keisler ordering

Let $\mathcal{U}$ be an ultrafilter over I and $f : I \to J$ be a function. Then, $\mathcal{U}$ induces an ultrafilter over J via f.

$$f[\mathcal{U}] := \{Y \subseteq J : f^{-1}(Y) \in \mathcal{U}\}$$

We verify that $f[\mathcal{U}]$ is indeed an ultrafilter.

(i) $\emptyset \notin f[\mathcal{U}]$ and $J \in f[\mathcal{U}]$: $f^{-1}(\emptyset) = \emptyset \notin \mathcal{U}$, so $\emptyset \notin f[\mathcal{U}]$; and $f^{-1}(J) = I \in \mathcal{U}$, so $J \in f[\mathcal{U}]$.

(ii) $Y_1, Y_2 \in f[\mathcal{U}] \Rightarrow Y_1 \cap Y_2 \in f[\mathcal{U}]$: $f^{-1}(Y_1 \cap Y_2) = f^{-1}(Y_1) \cap f^{-1}(Y_2) \in \mathcal{U}$, since $\mathcal{U}$ is closed under finite intersections. So $Y_1 \cap Y_2 \in f[\mathcal{U}]$.

(iii) $Y \in f[\mathcal{U}]$ and $Y \subseteq Y' \Rightarrow Y' \in f[\mathcal{U}]$: $f^{-1}(Y) \subseteq f^{-1}(Y')$, and since $f^{-1}(Y) \in \mathcal{U}$ and $\mathcal{U}$ is closed upward, $f^{-1}(Y') \in \mathcal{U}$. So $Y' \in f[\mathcal{U}]$.

(iv) For each $Y \subseteq J$, exactly one of $Y, J \setminus Y$ is in $f[\mathcal{U}]$: $f^{-1}(J \setminus Y) = I \setminus f^{-1}(Y)$. Since $\mathcal{U}$ is an ultrafilter, exactly one of $f^{-1}(Y), I \setminus f^{-1}(Y)$ is in $\mathcal{U}$, i.e. exactly one of $Y, J \setminus Y$ is in $f[\mathcal{U}]$.

Definition 8.1 (Rudin-Keisler ordering)**.** Let $\mathcal{U}, \mathcal{V}$ be ultrafilters over I, J respectively. Then $\mathcal{V} \leq_{\mathrm{RK}} \mathcal{U}$ if $\exists f : I \to J$ such that $\mathcal{V} = f[\mathcal{U}]$.

One can see that the order $\leq_{\mathrm{RK}}$ is transitive and reflexive by considering composition and the identity function respectively. So, it is a preorder on the collection of all ultrafilters. We say $\mathcal{U} \equiv_{\mathrm{RK}} \mathcal{V}$ if $\mathcal{U} \leq_{\mathrm{RK}} \mathcal{V}$ and $\mathcal{V} \leq_{\mathrm{RK}} \mathcal{U}$. Then $\equiv_{\mathrm{RK}}$ is an equivalence relation. We prove some simple facts about $\leq_{\mathrm{RK}}$-ordering.

Proposition 8.2. $\mathcal{V} \leq_{\mathrm{RK}} \mathcal{U} \Rightarrow \min\{|Y| : Y \in \mathcal{V}\} \leq \min\{|X| : X \in \mathcal{U}\}$.

Proof. Fix $f : I \to J$ with $\mathcal{V} = f[\mathcal{U}]$. For $X \in \mathcal{U}$, $f^{-1}(f[X]) \supseteq X$, so since $X \in \mathcal{U}$ and $\mathcal{U}$ is closed upward, $f^{-1}(f[X]) \in \mathcal{U}$, i.e. $f[X] \in f[\mathcal{U}] = \mathcal{V}$. Since $|f[X]| \leq |X|$, we get $\min\{|Y| : Y \in \mathcal{V}\} \leq |f[X]| \leq |X|$. As this holds for every $X \in \mathcal{U}$, taking X achieving $\min\{|X| : X \in \mathcal{U}\}$ gives $\min\{|Y| : Y \in \mathcal{V}\} \leq \min\{|X| : X \in \mathcal{U}\}$. □

Proposition 8.3. $\leq_{\mathrm{RK}}$ is upward directed.

Proof. Let $\mathcal{U}, \mathcal{V}$ be ultrafilters over I, J respectively, and let $\mathcal{W} := \mathcal{U} \times \mathcal{V}$, the product ultrafilter over $I \times J$ from **Proposition 5.5**. Let $\pi_I : I \times J \to I$ and $\pi_J : I \times J \to J$ be the coordinate projections. We show $\pi_I[\mathcal{W}] = \mathcal{U}$ and $\pi_J[\mathcal{W}] = \mathcal{V}$, which gives $\mathcal{U} \leq_{\mathrm{RK}} \mathcal{W}$ and $\mathcal{V} \leq_{\mathrm{RK}} \mathcal{W}$, i.e. $\mathcal{W}$ is a common upper bound.

For $Y \subseteq I$, $\pi_I^{-1}(Y) = Y \times J$, and $(Y \times J)_j = Y$ for every $j \in J$ (independent of j). So, by definition of the product ultrafilter, $Y \times J \in \mathcal{W} \Leftrightarrow \{j \in J : Y \in \mathcal{U}\} \in \mathcal{V}$; this set is J if $Y \in \mathcal{U}$ and $\emptyset$ otherwise, so it is in $\mathcal{V}$ exactly when $Y \in \mathcal{U}$. Hence $Y \in \pi_I[\mathcal{W}] \Leftrightarrow Y \times J \in \mathcal{W} \Leftrightarrow Y \in \mathcal{U}$, i.e. $\pi_I[\mathcal{W}] = \mathcal{U}$.

For $Z \subseteq J$, $\pi_J^{-1}(Z) = I \times Z$, and $(I \times Z)_j = I$ if $j \in Z$, and $(I \times Z)_j = \emptyset$ if $j \notin Z$. So $\{j : (I \times Z)_j \in \mathcal{U}\} = Z$ (using $I \in \mathcal{U}$ and $\emptyset \notin \mathcal{U}$), whence $I \times Z \in \mathcal{W} \Leftrightarrow Z \in \mathcal{V}$. Hence $\pi_J[\mathcal{W}] = \mathcal{V}$. □

$\leq_{\mathrm{RK}}$-ordering gives us a characterization of countably incomplete ultrafilters in terms of ultrafilters over ω. Recall that all non-principal ultrafilters over ω are countably incomplete (as shown in the example after **Definition 7.4**).

Proposition 8.4. $\mathcal{U}$ is countably incomplete $\Leftrightarrow \exists$ a non-principal ultrafilter $\mathcal{V}$ over ω st. $\mathcal{V} \leq_{\mathrm{RK}} \mathcal{U}$.

Proof. ($\Leftarrow$) Suppose $\mathcal{V} = f[\mathcal{U}]$ for some $f : I \to \omega$ and non-principal $\mathcal{V}$ over ω. Since $\mathcal{V}$ is non-principal, $\{n\} \notin \mathcal{V}$ for each n, so $\omega \setminus \{n\} \in \mathcal{V}$; hence every cofinite subset of ω is in $\mathcal{V}$ (being a finite intersection of such). In particular, the decreasing sets $V_n := \{n, n+1, n+2, \ldots\}$ are all in $\mathcal{V}$, and $\bigcap_n V_n = \emptyset$. Let $I_n := f^{-1}(V_n) \in \mathcal{U}$; then (I_n) is decreasing and $\bigcap_n I_n = f^{-1}\big(\bigcap_n V_n\big) = f^{-1}(\emptyset) = \emptyset$. So $\mathcal{U}$ is countably incomplete.

($\Rightarrow$) Suppose $\mathcal{U}$ is countably incomplete; fix $(I_n)_{n<\omega}$ in $\mathcal{U}$ with $I_n \supseteq I_{n+1}$, $\bigcap_n I_n = \emptyset$, and $I_0 = I$ (replacing I_n by $I \cap I_0 \cap \cdots \cap I_n$ if necessary). Define $f : I \to \omega$ by $f(i) := \max\{n : i \in I_n\}$ (well-defined as in earlier proofs), and let $\mathcal{V} := f[\mathcal{U}]$. If $\mathcal{V}$ were principal at some $m \in \omega$, then $\{m\} \in f[\mathcal{U}]$, i.e. $f^{-1}(\{m\}) = I_m \setminus I_{m+1} \in \mathcal{U}$. But $I_{m+1} \in \mathcal{U}$ too, and $(I_m \setminus I_{m+1}) \cap I_{m+1} = \emptyset$, contradicting that $\mathcal{U}$ is proper. So $\mathcal{V}$ is non-principal, and $\mathcal{V} \leq_{\mathrm{RK}} \mathcal{U}$ by construction. □

The next result characterizes exactly when $f[\mathcal{U}] \equiv_{\mathrm{RK}} \mathcal{U}$, purely in terms of the function f.

Theorem 8.5. Let $\mathcal{U}$ be an ultrafilter over I and $f : I \to J$ be a function. Then, $f[\mathcal{U}] \equiv_{\mathrm{RK}} \mathcal{U} \Leftrightarrow f$ is injective on some $X \in \mathcal{U}$.

Proof. Throughout, we use the easily-checked fact that for $f : I \to J$, $g : J \to K$, $(g \circ f)[\mathcal{U}] = g[f[\mathcal{U}]]$: indeed, for $Z \subseteq K$, $(g \circ f)^{-1}(Z) = f^{-1}(g^{-1}(Z))$, so $Z \in (g \circ f)[\mathcal{U}] \Leftrightarrow f^{-1}(g^{-1}(Z)) \in \mathcal{U} \Leftrightarrow g^{-1}(Z) \in f[\mathcal{U}] \Leftrightarrow Z \in g[f[\mathcal{U}]]$. We also use: if $X \in \mathcal{U}$ then for $A \subseteq I$, $A \in \mathcal{U} \Leftrightarrow A \cap X \in \mathcal{U}$ (the forward direction is closure under intersection; the converse is closure upward, as $A \cap X \subseteq A$).

($\Leftarrow$) Suppose f is injective on $X \in \mathcal{U}$. Fix $x_0 \in X$ and define $g : J \to I$ by $g(y) :=$ the unique $x \in X$ with $f(x) = y$, if $y \in f[X]$, and $g(y) := x_0$ otherwise. For $x \in X$, $g(f(x)) = x$ by construction. We claim $g[f[\mathcal{U}]] = \mathcal{U}$. For $Z \subseteq I$: $f^{-1}(g^{-1}(Z)) \cap X = \{x \in X : g(f(x)) \in Z\} = \{x \in X : x \in Z\} = Z \cap X$. Hence $f^{-1}(g^{-1}(Z)) \in \mathcal{U} \Leftrightarrow f^{-1}(g^{-1}(Z)) \cap X \in \mathcal{U} \Leftrightarrow Z \cap X \in \mathcal{U} \Leftrightarrow Z \in \mathcal{U}$, i.e. $Z \in g[f[\mathcal{U}]] \Leftrightarrow Z \in \mathcal{U}$. So $g[f[\mathcal{U}]] = \mathcal{U}$, giving $\mathcal{U} \leq_{\mathrm{RK}} f[\mathcal{U}]$. Since also $f[\mathcal{U}] \leq_{\mathrm{RK}} \mathcal{U}$ trivially (witnessed by f itself), we get $f[\mathcal{U}] \equiv_{\mathrm{RK}} \mathcal{U}$.

($\Rightarrow$) Suppose $f[\mathcal{U}] \equiv_{\mathrm{RK}} \mathcal{U}$, so (since $\mathcal{U} \leq_{\mathrm{RK}} f[\mathcal{U}]$) there is $g : J \to I$ with $g[f[\mathcal{U}]] = \mathcal{U}$. Let $h := g \circ f : I \to I$; then $h[\mathcal{U}] = g[f[\mathcal{U}]] = \mathcal{U}$. We show h (hence f, since $h = g \circ f$ injective on a set forces f injective there) is injective on some $X \in \mathcal{U}$; this suffices, since f injective on X follows from $h = g \circ f$ being injective on X.

Fix a well-ordering $<$ of I, and let $B := \{i \in I : |h^{-1}(\{h(i)\})| \geq 2\}$. If $B \notin \mathcal{U}$, then $X := I \setminus B \in \mathcal{U}$ and h is injective on X by definition of B, as desired. So suppose $B \in \mathcal{U}$. For $i \in B$, let $c(i) := 0$ if i is the $<$-least element of its fibre $h^{-1}(\{h(i)\})$, and $c(i) := 1$ otherwise. Then $c^{-1}(0)$ and $c^{-1}(1)$ partition $B \in \mathcal{U}$, so exactly one of them is in $\mathcal{U}$ (if neither were, their complements in I would both be in $\mathcal{U}$, forcing their intersection $I \setminus B \in \mathcal{U}$, contradicting $B \in \mathcal{U}$; and they cannot both be in $\mathcal{U}$ as they are disjoint).

Note that $h^{-1}(c^{-1}(0)) = B$: indeed, $c^{-1}(0)$ consists of exactly one representative from each fibre of size ≥ 2, so $h[c^{-1}(0)]$ is exactly the set of values attained on such fibres, and $h^{-1}(h[c^{-1}(0)])$ is the union of those (complete) fibres, i.e. B. Since $h[\mathcal{U}] = \mathcal{U}$, we have $c^{-1}(0) \in \mathcal{U} \Leftrightarrow h^{-1}(c^{-1}(0)) = B \in \mathcal{U}$, and the latter holds by assumption. So $c^{-1}(0) \in \mathcal{U}$ (this also rules out the case $c^{-1}(1) \in \mathcal{U}$, since the two are disjoint). Finally, h is injective on $X := c^{-1}(0)$: distinct elements of $c^{-1}(0)$ are the least elements of distinct fibres of h, hence have distinct h-values. □

The next result shows that ultraproducts over $\leq_{\mathrm{RK}}$-lesser ultrafilters are elementarily embeddable in ultraproducts over $\leq_{\mathrm{RK}}$-higher ultrafilters. More precisely, we have the following result.

Theorem 8.6. Let $\mathcal{U}, \mathcal{V}$ be ultrafilters over I, J respectively. Then we have the following.

(i) $\mathcal{V} \leq_{\mathrm{RK}} \mathcal{U} \Leftrightarrow$ for all $\mathcal{L}$-structures $\mathcal{M}$, $\prod_{\mathcal{V}} \mathcal{M} \preceq \prod_{\mathcal{U}} \mathcal{M}$.

(ii) $\mathcal{V} \equiv_{\mathrm{RK}} \mathcal{U} \Leftrightarrow$ for all $\mathcal{L}$-structures $\mathcal{M}$, $\prod_{\mathcal{V}} \mathcal{M} \cong \prod_{\mathcal{U}} \mathcal{M}$.

Proof. **(i),** $(\Rightarrow)$**.** Suppose $\mathcal{V} = f[\mathcal{U}]$ for some $f : I \to J$. Fix an $\mathcal{L}$-structure $\mathcal{M}$ with universe M, and define $e : \prod_{\mathcal{V}} \mathcal{M} \to \prod_{\mathcal{U}} \mathcal{M}$ by $e([k]_{\mathcal{V}}) := [k \circ f]_{\mathcal{U}}$ for $k : J \to M$.

Well-defined. If $k =_{\mathcal{V}} k'$, i.e. $\{j : k(j) = k'(j)\} \in \mathcal{V} = f[\mathcal{U}]$, then $f^{-1}(\{j : k(j) = k'(j)\}) = \{i : k(f(i)) = k'(f(i))\} \in \mathcal{U}$, i.e. $k \circ f =_{\mathcal{U}} k' \circ f$.

Elementary. Fix an $\mathcal{L}$-formula $\varphi(x)$ and $k \in \prod_{\mathcal{V}} M$. By Łoś' theorem (applied to the constant families $\mathcal{M}$ over J w.r.t. $\mathcal{V}$, and over I w.r.t. $\mathcal{U}$),

$$\prod_{\mathcal{V}} \mathcal{M} \models \varphi([k]_{\mathcal{V}}) \Leftrightarrow \{j \in J : \mathcal{M} \models \varphi(k(j))\} \in \mathcal{V} = f[\mathcal{U}] \Leftrightarrow f^{-1}\big(\{j : \mathcal{M} \models \varphi(k(j))\}\big) \in \mathcal{U}.$$

Since $f^{-1}\big(\{j : \mathcal{M} \models \varphi(k(j))\}\big) = \{i : \mathcal{M} \models \varphi(k(f(i)))\} = \{i : \mathcal{M} \models \varphi((k \circ f)(i))\}$, this last condition is, again by Łoś' theorem, equivalent to $\prod_{\mathcal{U}} \mathcal{M} \models \varphi([k \circ f]_{\mathcal{U}}) = \varphi(e([k]_{\mathcal{V}}))$. So e preserves and reflects every $\mathcal{L}$-formula; in particular it is injective (take $\varphi(x, y) := \neg(x = y)$), so $e : \prod_{\mathcal{V}} \mathcal{M} \prec \prod_{\mathcal{U}} \mathcal{M}$.

(i), $(\Leftarrow)$**.** Suppose that for every $\mathcal{L}$-structure $\mathcal{M}$, $\prod_{\mathcal{V}} \mathcal{M} \preceq \prod_{\mathcal{U}} \mathcal{M}$. Let $\mathcal{L} := \{P_Z : Z \subseteq J\}$, one unary relation symbol for each subset of J, and let $\mathcal{A}$ be the $\mathcal{L}$-structure with universe J and $P_Z^{\mathcal{A}} := Z$ for each $Z \subseteq J$. By hypothesis, there is an elementary embedding $e : \prod_{\mathcal{V}} \mathcal{A} \prec \prod_{\mathcal{U}} \mathcal{A}$. The universe of $\prod_{\mathcal{V}} \mathcal{A}$ is $\prod_{\mathcal{V}} J$; let $\iota := \mathrm{id}_J$ (viewed as an element of $\prod_{j \in J} J$), and write $e([\iota]_{\mathcal{V}}) =: [f]_{\mathcal{U}}$ for some $f : I \to J$.

Fix $Z \subseteq J$. Since $\mathcal{A} \models P_Z(\iota(j')) \Leftrightarrow j' \in Z$, Łoś' theorem (w.r.t. $\mathcal{V}$) gives $\prod_{\mathcal{V}} \mathcal{A} \models P_Z([\iota]_{\mathcal{V}}) \Leftrightarrow Z \in \mathcal{V}$. By elementarity of e and Łoś' theorem again (w.r.t. $\mathcal{U}$, noting $\mathcal{A} \models P_Z(f(i)) \Leftrightarrow f(i) \in Z$),

$$\prod_{\mathcal{V}} \mathcal{A} \models P_Z([\iota]_{\mathcal{V}}) \Leftrightarrow \prod_{\mathcal{U}} \mathcal{A} \models P_Z([f]_{\mathcal{U}}) \Leftrightarrow \{i \in I : f(i) \in Z\} = f^{-1}(Z) \in \mathcal{U}.$$

So $Z \in \mathcal{V} \Leftrightarrow f^{-1}(Z) \in \mathcal{U}$ for every $Z \subseteq J$, i.e. $f[\mathcal{U}] = \mathcal{V}$, giving $\mathcal{V} \leq_{\mathrm{RK}} \mathcal{U}$.

(ii), $(\Rightarrow)$**.** Suppose $\mathcal{V} \equiv_{\mathrm{RK}} \mathcal{U}$, i.e. $\mathcal{V} = f[\mathcal{U}]$ and $\mathcal{U} = g[\mathcal{V}]$ for some $f : I \to J$, $g : J \to I$. Let $h := g \circ f$; then $h[\mathcal{U}] = g[f[\mathcal{U}]] = g[\mathcal{V}] = \mathcal{U}$ (using the composition fact from the proof of the previous theorem). By that theorem, h is injective on some $X \in \mathcal{U}$, and since $h = g \circ f$ is injective on X, so is f.

Fix an $\mathcal{L}$-structure $\mathcal{M}$ with universe M, and let $e : \prod_{\mathcal{V}} \mathcal{M} \prec \prod_{\mathcal{U}} \mathcal{M}$, $[k]_{\mathcal{V}} \mapsto [k \circ f]_{\mathcal{U}}$, be the elementary embedding constructed in part (i). We show e is onto. Let $[l]_{\mathcal{U}} \in \prod_{\mathcal{U}} M$. Fix $x_0 \in X$ and define $k : J \to M$ by $k(j) :=$ the value $l(x)$ for the unique $x \in X$ with $f(x) = j$, if $j \in f[X]$, and $k(j) := l(x_0)$ otherwise (using injectivity of f on X). For $x \in X$, $k(f(x)) = l(x)$; since $X \in \mathcal{U}$, $[k \circ f]_{\mathcal{U}} = [l]_{\mathcal{U}}$, i.e. $e([k]_{\mathcal{V}}) = [l]_{\mathcal{U}}$. So e is a bijective elementary embedding, hence an isomorphism $\prod_{\mathcal{V}} \mathcal{M} \cong \prod_{\mathcal{U}} \mathcal{M}$.

(ii), $(\Leftarrow)$**.** If $\prod_{\mathcal{V}} \mathcal{M} \cong \prod_{\mathcal{U}} \mathcal{M}$ for every $\mathcal{L}$-structure $\mathcal{M}$, then in particular $\prod_{\mathcal{V}} \mathcal{M} \preceq \prod_{\mathcal{U}} \mathcal{M}$ for every $\mathcal{M}$ (an isomorphism is an elementary embedding), giving $\mathcal{V} \leq_{\mathrm{RK}} \mathcal{U}$ by part (i); and likewise, the inverse isomorphisms give $\prod_{\mathcal{U}} \mathcal{M} \preceq \prod_{\mathcal{V}} \mathcal{M}$ for every $\mathcal{M}$, giving $\mathcal{U} \leq_{\mathrm{RK}} \mathcal{V}$. Hence $\mathcal{V} \equiv_{\mathrm{RK}} \mathcal{U}$. □

9 Regular ultrafilters

Before moving on to the class of good ultrafilters (which yield ultraproducts with higher levels of saturation), it behooves us to first consider the class of regular ultrafilters. While the definition of good ultrafilters is somewhat complicated and their existence is quite cumbersome to prove, the class of regular ultrafilters is quite easy to define and their existence is a simple exercise. We will see that ultrapowers (not arbitrary ultraproducts) with respect to regular ultrafilters are highly universal.

Proposition 9.1. $\mathcal{U}$ is an ultrafilter over an index set I. Then $\mathcal{U}$ is countably incomplete if and only if there is $E \subseteq \mathcal{U}$, $|E| = \aleph_0$, such that each element of I belongs to only finitely many elements of E.

Proof. ($\Rightarrow$) Fix $(I_n)_{n<\omega}$ with $I_n \in \mathcal{U}$, $I_n \supseteq I_{n+1}$, and $\bigcap_n I_n = \emptyset$ (as in earlier proofs). Let $E := \{I_n : n < \omega\}$. For $i \in I$, since $\bigcap_n I_n = \emptyset$, $i \notin I_n$ for some n, and since (I_n) is decreasing, $i \notin I_m$ for all $m \geq n$ either. So i belongs to only the (at most) finitely many sets $I_0, \ldots, I_{n-1}$.

($\Leftarrow$) Suppose $E = \{E_n : n < \omega\} \subseteq \mathcal{U}$ with each $i \in I$ in only finitely many E_n. Let $I_n := \bigcap_{k \leq n} E_k \in \mathcal{U}$; this is decreasing. For $i \in I$, let N be largest such that $i \in E_N$ (exists since i is in only finitely many E_n); then $i \notin E_n$ for $n > N$, so $i \notin I_n$ for $n > N$. Hence $\bigcap_n I_n = \emptyset$, so $\mathcal{U}$ is countably incomplete. □

We strengthen the notion of countably incomplete ultrafilter by demanding that there exists a set E as above that has cardinality larger than $\aleph_0$:

Definition 9.2. If κ is a cardinal, then an ultrafilter $\mathcal{U}$ on I is **κ-regular** if there is $E \subseteq \mathcal{U}$, $|E| = \kappa$, such that each element of I belongs to only finitely many elements of E. We call such an E a κ-regularizing set for $\mathcal{U}$.

The previous proposition can be rephrased as:

Lemma 9.3. An ultrafilter is countably incomplete if and only if it is $\aleph_0$-regular.

Proposition 9.4. Suppose $\mathcal{U}$ is a κ-regular ultrafilter on an index set I, with κ infinite. Then:

1. $\mathcal{U}$ is non-principal.
2. If $\lambda < \kappa$, then $\mathcal{U}$ is also λ-regular.
3. $\kappa \leq |I|$.

Proof. Let $E = \{E_\alpha : \alpha < \kappa\}$ be a κ-regularizing set.

(1) Suppose $\mathcal{U}$ is principal at $i_0 \in I$. Then every $A \in \mathcal{U}$ contains i_0; in particular $i_0 \in E_\alpha$ for every $\alpha < \kappa$. Since κ is infinite, i_0 belongs to infinitely many elements of E, contradicting regularity. So $\mathcal{U}$ is non-principal.

(2) Any λ-sized subset of E is still a subset of $\mathcal{U}$ in which each $i \in I$ belongs to only finitely many elements (a fortiori, since it is a subcollection of E), so it witnesses λ-regularity.

(3) Each E_α is non-empty (as $\emptyset \notin \mathcal{U}$), so for each $\alpha < \kappa$ we may fix $i_\alpha \in E_\alpha$. The map $i \mapsto \{\alpha < \kappa : i \in E_\alpha\}$ assigns to each $i \in I$ a finite subset of κ, and every $\alpha < \kappa$ lies in the finite set assigned to i_α. So $\kappa = \left|\bigcup_{i \in I} \{\alpha : i \in E_\alpha\}\right| \leq |I| \cdot \aleph_0$. By (1) and the fact that only infinite index sets admit non-principal ultrafilters, I is infinite, so $|I| \cdot \aleph_0 = |I|$, giving $\kappa \leq |I|$. □

By the previous proposition, regular ultrafilters are "maximally" regular in the following sense.

Definition 9.5. An ultrafilter $\mathcal{U}$ on an index set I is called **regular** if it is $|I|$-regular.

The following reformulation of κ-regularity is often useful.

Proposition 9.6. Suppose $\mathcal{U}$ is an ultrafilter on I and κ is an infinite cardinal. Then $\mathcal{U}$ is κ-regular if and only if there is $f : I \to \mathcal{P}_f(\kappa)$ such that, for each $\alpha < \kappa$, $\{i \in I : \alpha \in f(i)\} \in \mathcal{U}$.

Proof. ($\Rightarrow$) Let $\{E_\alpha : \alpha < \kappa\}$ be κ-regularizing. Define $f(i) := \{\alpha < \kappa : i \in E_\alpha\}$, a finite subset of κ by regularity. Then $\{i : \alpha \in f(i)\} = \{i : i \in E_\alpha\} = E_\alpha \in \mathcal{U}$.

($\Leftarrow$) Given such an f, let $E_\alpha := \{i \in I : \alpha \in f(i)\} \in \mathcal{U}$ for $\alpha < \kappa$. For $i \in I$, $\{\alpha < \kappa : i \in E_\alpha\} = \{\alpha : \alpha \in f(i)\} = f(i)$, which is finite. So $E := \{E_\alpha : \alpha < \kappa\}$ witnesses κ-regularity of $\mathcal{U}$. □

Proposition 9.7. Suppose $\mathcal{U}, \mathcal{V}$ are ultrafilters, over I, J respectively, with $\mathcal{U} \leq_{\mathrm{RK}} \mathcal{V}$. If $\mathcal{U}$ is κ-regular, then so is $\mathcal{V}$. In particular, if $\mathcal{U}$ or $\mathcal{V}$ (or both) is κ-regular, then so is $\mathcal{U} \times \mathcal{V}$.

Proof. Since $\mathcal{U} \leq_{\mathrm{RK}} \mathcal{V}$, fix $f : J \to I$ with $\mathcal{U} = f[\mathcal{V}]$. Let $\{E_\alpha : \alpha < \kappa\} \subseteq \mathcal{U}$ be κ-regularizing. For each α, $f^{-1}(E_\alpha) \in \mathcal{V}$ (by definition of $f[\mathcal{V}] = \mathcal{U} \ni E_\alpha$). For $j \in J$: $j \in f^{-1}(E_\alpha) \Leftrightarrow f(j) \in E_\alpha$; since $f(j) \in I$ belongs to only finitely many E_α, j belongs to only finitely many $f^{-1}(E_\alpha)$. So $\{f^{-1}(E_\alpha) : \alpha < \kappa\}$ is κ-regularizing for $\mathcal{V}$.

For the final claim, recall from the section on the Rudin-Keisler ordering that $\mathcal{U} \leq_{\mathrm{RK}} \mathcal{U} \times \mathcal{V}$ and $\mathcal{V} \leq_{\mathrm{RK}} \mathcal{U} \times \mathcal{V}$ (via the coordinate projections). Applying the result just proved with $\mathcal{U} \times \mathcal{V}$ in place of $\mathcal{V}$ shows that κ-regularity of either $\mathcal{U}$ or $\mathcal{V}$ propagates to $\mathcal{U} \times \mathcal{V}$. □

We now show that regular ultrafilters exist.

Proposition 9.8. For each infinite cardinal κ, there is a regular ultrafilter $\mathcal{U}$ on κ.

Proof. It suffices to find a regular ultrafilter on some set of size κ; we use $I := \mathcal{P}_f(\kappa)$. For $\alpha < \kappa$, let $\hat{\alpha} := \{u \in I : \alpha \in u\}$. Note $\alpha \neq \beta \Rightarrow \hat{\alpha} \neq \hat{\beta}$, since $\{\alpha\} \in \hat{\alpha} \setminus \hat{\beta}$; so $E := \{\hat{\alpha} : \alpha < \kappa\}$ has $|E| = \kappa$. Also, E has the finite intersection property: $\{\alpha_1, \ldots, \alpha_n\} \in \bigcap_{k=1}^n \hat{\alpha}_k$. Let $\mathcal{U}$ be an ultrafilter on I containing E (exists by extending the filter generated by E). Then E is a κ-regularizing set for $\mathcal{U}$: for $u \in I$ (a finite subset of κ), $u \in \hat{\alpha} \Leftrightarrow \alpha \in u$, so $\{\alpha < \kappa : u \in \hat{\alpha}\} = u$, which is finite. Since $|I| = \kappa$, $\mathcal{U}$ is regular. □

Ultrapowers with respect to regular ultrafilters have predictable cardinalities.

Theorem 9.9. Suppose $\mathcal{U}$ is a regular ultrafilter on κ and M is infinite. Then $|M^{\mathcal{U}}| = |M|^\kappa$.

Proof. It suffices to prove $|M|^\kappa \leq |M^{\mathcal{U}}|$ (the other inequality always holds). Let $E \subseteq \mathcal{U}$ be a κ-regularizing set for $\mathcal{U}$. Let $N := M^{<\omega}$. It suffices to find an injection $\rho : M^E \to N^{\mathcal{U}}$. For each $g \in M^E$, define $g' : I \to N$ by $g'(i) := (g(A_1), \ldots, g(A_n))$, where $A_1, \ldots, A_n$ enumerate (with respect to some fixed ordering of E) all elements of E containing i (finitely many, by regularity). We define $\rho(g) := [g']_{\mathcal{U}}$. It remains to show ρ is injective. Suppose $g \neq h$ and take $A \in E$ such that $g(A) \neq h(A)$. If $i \in A$, then A is one of the (finitely many) elements of E enumerated in $g'(i)$ and $h'(i)$ at the same position, so $g'(i) \neq h'(i)$. Since $A \in \mathcal{U}$, $\rho(g) \neq \rho(h)$. □

We now state the connection between regularity and universality.

Theorem 9.10. Suppose $\mathcal{U}$ is an ultrafilter on an index set I. Then $\mathcal{U}$ is κ-regular if and only if, whenever $|\mathcal{L}| \leq \kappa$ and $\mathcal{M}$ is an $\mathcal{L}$-structure, we have that $\mathcal{M}^{\mathcal{U}}$ is κ^+-universal.

Proof. First suppose $\mathcal{U}$ is κ-regular, $|\mathcal{L}| \leq \kappa$, and $\mathcal{M}$ is an $\mathcal{L}$-structure. Set $\mathcal{N} := \mathcal{M}^{\mathcal{U}}$. We use the criterion established above to show $\mathcal{N}$ is κ^+-universal. Let Σ be a set of $\mathcal{L}(\kappa)$-sentences such that $\mathrm{Th}(\mathcal{N}) \cup \Sigma$ is satisfiable. Let $E \subseteq \mathcal{U}$ be a κ-regularizing set for $\mathcal{U}$ and let $h : \Sigma \to E$ be any injection. Let $\Sigma(i) := \{\sigma \in \Sigma : i \in h(\sigma)\}$. Since E is regularizing and h is injective, $\Sigma(i)$ is finite for each $i \in I$. Since $\mathrm{Th}(\mathcal{N}) = \mathrm{Th}(\mathcal{M})$ (as $\mathcal{M} \equiv \mathcal{N}$), each finite $\Sigma(i)$ is consistent with $\mathrm{Th}(\mathcal{M})$, so there is $a(i) \in M^\kappa$ with $(\mathcal{M}; a(i)) \models \Sigma(i)$ (only finitely many constants are mentioned, so most of the choice of $a(i)$ is irrelevant). Set $a \in N^\kappa$ by $a(\alpha) := [i \mapsto a(i)(\alpha)]_{\mathcal{U}}$. We claim Σ is realized in $(\mathcal{N}; a)$: fix $\sigma \in \Sigma$; then $h(\sigma) \in \mathcal{U}$, and for $i \in h(\sigma)$ we have $\sigma \in \Sigma(i)$, so $(\mathcal{M}; a(i)) \models \sigma$. By Łoś' theorem (applied to the constant family $(\mathcal{M}; a(i))_{i \in I}$), $(\mathcal{N}; a) \models \sigma$. So Σ is satisfied in $(\mathcal{N}; a)$, giving κ^+-universality of $\mathcal{N}$ by the earlier criterion.

Conversely, suppose that whenever $|\mathcal{L}| \leq \kappa$ and $\mathcal{M}$ is an $\mathcal{L}$-structure, $\mathcal{M}^{\mathcal{U}}$ is κ^+-universal. Consider the language $\mathcal{L} = \{R\} \cup \{d_\alpha : \alpha < \kappa\}$ with R binary and each d_α a constant, and let $\mathcal{M}$ be the $\mathcal{L}$-structure with universe $\mathcal{P}_f(\kappa)$, interpreting R as $\subseteq$ and d_α as $\{\alpha\}$. By assumption, $\mathcal{M}^{\mathcal{U}}$ is κ^+-universal. Let $\Sigma := \{R(d_\alpha, c_0) : \alpha < \kappa\}$, a set of $\mathcal{L}(\kappa)$-sentences (mentioning only the single new constant c_0). Every finite subset of Σ is satisfiable in a suitable expansion of $\mathcal{M}$ (take c_0 to be a finite subset of κ containing the finitely many α's mentioned), so $\mathrm{Th}(\mathcal{M}) \cup \Sigma$ is finitely satisfiable, hence satisfiable by the Compactness Theorem. Since $\mathcal{M}^{\mathcal{U}}$ is κ^+-universal, there is $b \in M^{\mathcal{U}}$ such that Σ is satisfied in $(\mathcal{M}^{\mathcal{U}}; [b]_{\mathcal{U}})$, where $b : I \to \mathcal{P}_f(\kappa)$.

For each $\alpha < \kappa$, let $A_\alpha := \{i \in I : \alpha \in b(i)\}$. By Łoś' theorem, $(\mathcal{M}^{\mathcal{U}}; [b]_{\mathcal{U}}) \models R(d_\alpha, c_0)$ means exactly $A_\alpha \in \mathcal{U}$. Let $E := \{A_\alpha : \alpha < \kappa\}$. For $i \in I$: $i \in A_\alpha \Leftrightarrow \alpha \in b(i)$, and $b(i) \in \mathcal{P}_f(\kappa)$ is finite, so i belongs to only finitely many A_α. So E is a κ-regularizing set for $\mathcal{U}$. □

The following corollary is immediate.

Corollary 9.11 (Frayne)**.** $\mathcal{L}$-structures $\mathcal{M}$ and $\mathcal{N}$ are elementarily equivalent if and only if $\mathcal{M}$ is elementarily embeddable in an ultrapower of $\mathcal{N}$.

Proof. ($\Leftarrow$) Ultrapowers are elementary extensions (via the diagonal embedding), so if $\mathcal{M}$ embeds elementarily into $\mathcal{N}^{\mathcal{U}} \equiv \mathcal{N}$, then $\mathcal{M} \equiv \mathcal{N}^{\mathcal{U}} \equiv \mathcal{N}$.

($\Rightarrow$) Suppose $\mathcal{M} \equiv \mathcal{N}$. Let $\kappa := \max(|M|, |\mathcal{L}|)$ and fix a regular (hence κ-regular, as it is κ'-regular for $\kappa' := |I| \geq \kappa$ after choosing $|I| = \kappa$) ultrafilter $\mathcal{U}$ on a set of size κ. By the previous theorem, $\mathcal{N}^{\mathcal{U}}$ is κ^+-universal. Since $\mathcal{M} \equiv \mathcal{N} \equiv \mathcal{N}^{\mathcal{U}}$ and $|M| \leq \kappa$, κ^+-universality gives an elementary embedding $\mathcal{M} \prec \mathcal{N}^{\mathcal{U}}$. □

We can use the preceding corollary to establish a fact about chains of ultrapowers. Given an $\mathcal{L}$-structure $\mathcal{M}$, an **ultrapower chain** over $\mathcal{M}$ is a chain $\mathcal{M} = \mathcal{M}_0 \prec \mathcal{M}_1 \prec \mathcal{M}_2 \prec \cdots$ where each $\mathcal{M}_{n+1}$ is (isomorphic to) an ultrapower of $\mathcal{M}_n$; its **limit** is $\mathcal{M}_\infty := \bigcup_n \mathcal{M}_n$.

Corollary 9.12. $\mathcal{L}$-structures $\mathcal{M}$ and $\mathcal{N}$ are elementarily equivalent if and only if there are ultrapower chains $(\mathcal{M}_n)_{n \in \mathbb{N}}$ and $(\mathcal{N}_n)_{n \in \mathbb{N}}$ over $\mathcal{M}$ and $\mathcal{N}$ respectively, whose limits $\mathcal{M}_\infty$ and $\mathcal{N}_\infty$ are isomorphic.

Proof sketch. The backward direction is immediate, since $\mathcal{M}$ and $\mathcal{N}$ are elementary substructures of $\mathcal{M}_\infty$ and $\mathcal{N}_\infty$ respectively. For the forward direction, suppose $\mathcal{M} \equiv \mathcal{N}$. By the previous corollary, there is an ultrapower $\mathcal{N}_1$ of $\mathcal{N}$ and an elementary embedding $f_1 : \mathcal{M} \to \mathcal{N}_1$. Let $\mathcal{L}'$ be the expansion of $\mathcal{L}$ by constants for all elements of M, and let $\mathcal{M}'$ and $\mathcal{N}_1'$ be the corresponding $\mathcal{L}'$-expansions of $\mathcal{M}$ and $\mathcal{N}_1$ (interpreting the new constant for $a \in M$ as a and $f_1(a)$ respectively). Since f_1 is elementary, $\mathcal{M}' \equiv \mathcal{N}_1'$. Applying the previous corollary again, there is an ultrapower $\mathcal{M}_1'$ of $\mathcal{M}'$ into which $\mathcal{N}_1'$ elementarily embeds; let $\mathcal{M}_1$ denote the $\mathcal{L}$-reduct of $\mathcal{M}_1'$, which is an ultrapower extension of $\mathcal{M}$, and let $g_1 : \mathcal{N}_1 \to \mathcal{M}_1$ be the induced elementary embedding, which (by design) extends f_1^{-1} on $f_1[M]$, i.e. $g_1 \circ f_1$ is the inclusion $M \hookrightarrow M_1$. Continuing in this back-and-forth manner, we build ultrapower chains $(\mathcal{M}_n)$, $(\mathcal{N}_n)$ and elementary embeddings $f_n : \mathcal{M}_n \to \mathcal{N}_{n+1}$, $g_n : \mathcal{N}_n \to \mathcal{M}_{n+1}$ that extend each other appropriately. Setting $f := \bigcup_n f_n$ and $g := \bigcup_n g_n$, f and g are inverse isomorphisms between $\mathcal{M}_\infty$ and $\mathcal{N}_\infty$. □

The following is one of the fundamental facts about regular ultrafilters; it is the driving force behind Keisler's order, discussed in a later section.

Theorem 9.13 (Keisler)**.** Suppose $\mathcal{U}$ is a κ-regular ultrafilter on the set I. Suppose $|\mathcal{L}| \leq \kappa$ and $\mathcal{M}$ and $\mathcal{N}$ are $\mathcal{L}$-structures with $\mathcal{M} \equiv \mathcal{N}$. Then $\mathcal{M}^{\mathcal{U}}$ is κ^+-saturated if and only if $\mathcal{N}^{\mathcal{U}}$ is κ^+-saturated.

Proof. Suppose $\mathcal{M}^{\mathcal{U}}$ is κ^+-saturated and let $E \subseteq \mathcal{U}$ be a κ-regularizing subset of $\mathcal{U}$. Suppose $\Sigma(x)$ is a set of $\mathcal{L}(\kappa)$-formulae that is finitely satisfiable in $(\mathcal{N}^{\mathcal{U}}; b)$. We show $\Sigma(x)$ is satisfiable in $(\mathcal{N}^{\mathcal{U}}; b)$.

Let $h : \Sigma \to E$ be an injection and set $\Sigma(i) := \{\sigma \in \Sigma : i \in h(\sigma)\}$; as before, each $\Sigma(i)$ is finite. Let $\Gamma(i)$ be the finite set of sentences of the form $\exists x \bigwedge_{\sigma \in u} \sigma$, where u is a non-empty subset of $\Sigma(i)$. Since $\mathcal{M} \equiv \mathcal{N}$, there is $a(i) \in M^\kappa$ such that

$$(\dagger) \qquad \mathrm{Th}(\mathcal{M}, a(i)) \cap \Gamma(i) = \mathrm{Th}(\mathcal{N}, b(i)) \cap \Gamma(i).$$

(Indeed, since $\Sigma(i)$ is finitely satisfiable in $(\mathcal{N}^{\mathcal{U}}; b)$, hence in $(\mathcal{N}; b(i))$ for $\mathcal{U}$-almost all i, every sentence in $\Gamma(i)$ true of $(\mathcal{N}; b(i))$ is consistent with $\mathrm{Th}(\mathcal{N}) = \mathrm{Th}(\mathcal{M})$, so by finite satisfiability and elementary equivalence we may choose $a(i)$ realizing exactly the same sentences of $\Gamma(i)$.)

Define $a \in (M^{\mathcal{U}})^\kappa$ by $a(\alpha) := [i \mapsto a(i)(\alpha)]_{\mathcal{U}}$. We claim $\Sigma(x)$ is finitely satisfiable in $(\mathcal{M}^{\mathcal{U}}; a)$. Fix $\sigma_1, \ldots, \sigma_n \in \Sigma$ and let $\tau := \exists x \bigwedge_{j=1}^n \sigma_j$. If $i \in \bigcap_{j=1}^n h(\sigma_j)$, then $\tau \in \Gamma(i)$. Since Σ is finitely satisfiable in $(\mathcal{N}^{\mathcal{U}}; b)$, we have $\tau \in \mathrm{Th}(\mathcal{N}^{\mathcal{U}}, b)$, so for $\mathcal{U}$-almost all i, $(\mathcal{N}; b(i)) \models \tau$. It follows that for $\mathcal{U}$-almost all i, $(\mathcal{M}; a(i)) \models \tau$ (by $(\dagger)$), whence $(\mathcal{M}^{\mathcal{U}}; a) \models \tau$, as desired.

Since $\mathcal{M}^{\mathcal{U}}$ is κ^+-saturated, Σ is realized in $(\mathcal{M}^{\mathcal{U}}; a)$, say by $[c]_{\mathcal{U}}$. For $i \in I$, let $\Psi(i) := \{\sigma \in \Sigma(i) : (\mathcal{M}; a(i)) \models \sigma(c(i))\}$. By $(\dagger)$, there is $d(i) \in N$ such that $(\mathcal{N}; b(i)) \models \sigma(d(i))$ for each $\sigma \in \Psi(i)$.

We conclude by showing $[d]_{\mathcal{U}}$ realizes Σ in $(\mathcal{N}^{\mathcal{U}}; b)$. Fix $\sigma \in \Sigma$. For $i \in h(\sigma)$, we have $\sigma \in \Sigma(i)$; since $(\mathcal{M}^{\mathcal{U}}; a) \models \sigma([c]_{\mathcal{U}})$, we have $(\mathcal{M}; a(i)) \models \sigma(c(i))$ for $\mathcal{U}$-almost all i. It follows that, for $\mathcal{U}$-almost all such i, $\sigma \in \Psi(i)$, whence $(\mathcal{N}; b(i)) \models \sigma(d(i))$. Since $h(\sigma) \in \mathcal{U}$, by Łoś' theorem $(\mathcal{N}^{\mathcal{U}}; b) \models \sigma([d]_{\mathcal{U}})$, as desired. □

Returning to the universality theorem above: if $\mathcal{U}$ is regular on κ, $|\mathcal{L}| \leq \kappa$, and $\mathcal{M}$ has $|M| \leq \kappa$, then $|M^{\mathcal{U}}| = |M|^\kappa \geq \kappa^+$, so in principle $\mathcal{M}^{\mathcal{U}}$ could even be κ^{++}-universal, not merely κ^+-universal as guaranteed above. (If the Generalized Continuum Hypothesis holds at κ, $|M^{\mathcal{U}}| = \kappa^+$ exactly.) The κ^+-good ultrafilters studied in the next sections yield κ^+-saturated, and hence κ^{++}-universal, ultrapowers.

Before moving on, we record two further applications of the universality theorem. Given an $\mathcal{L}$-structure $\mathcal{M}$, define the **universal theory** of $\mathcal{M}$, denoted $\mathrm{Th}_\forall(\mathcal{M})$, to be the set of universal $\mathcal{L}$-sentences σ such that $\mathcal{M} \models \sigma$.

Proposition 9.14. Given $\mathcal{L}$-structures $\mathcal{M}$ and $\mathcal{N}$, the following are equivalent:

1. $\mathcal{M} \models \mathrm{Th}_\forall(\mathcal{N})$.
2. For every κ-regular ultrafilter $\mathcal{U}$, where $\kappa = \max(|M|, |\mathcal{L}|)$, $\mathcal{M}$ embeds into $\mathcal{N}^{\mathcal{U}}$.
3. $\mathcal{M}$ embeds into some ultrapower $\mathcal{N}^{\mathcal{U}}$ of $\mathcal{N}$.

Proof. (1) implies (2): Let $\kappa = \max(|M|, |\mathcal{L}|)$ and let $\mathcal{U}$ be κ-regular. Let $\mathrm{Diag}(\mathcal{M})$ be the atomic diagram of $\mathcal{M}$ (the set of quantifier-free $\mathcal{L}_M$-sentences true in $\mathcal{M}$), viewed as a set of $\mathcal{L}(\kappa)$-sentences (one new constant per element of M, using $|M| \leq \kappa$). By the Diagram Lemma, it suffices to show $\mathrm{Th}(\mathcal{N}) \cup \mathrm{Diag}(\mathcal{M})$ is satisfiable, as $\mathcal{N}^{\mathcal{U}}$ is then κ^+-universal by the previous theorem, and so (by the earlier criterion for universality) realizes $\mathrm{Diag}(\mathcal{M})$ in some $(\mathcal{N}^{\mathcal{U}}; a)$, yielding the desired embedding.

By compactness, it suffices to show every finite subset of $\mathrm{Diag}(\mathcal{M})$ is consistent with $\mathrm{Th}(\mathcal{N})$. Given a finite conjunction $\theta(\bar{x})$ of quantifier-free formulae true of a tuple $\bar{a}$ in $\mathcal{M}$, we have $\mathcal{M} \models \exists \bar{x}\, \theta(\bar{x})$. If $\mathrm{Th}(\mathcal{N}) \cup$

$\{\exists \bar{x}\, \theta(\bar{x})\}$ were inconsistent, then (as $\mathrm{Th}(\mathcal{N})$ is complete) $\mathcal{N} \models \forall \bar{x}\, \neg\theta(\bar{x})$, a universal sentence, so by (1), $\mathcal{M} \models \forall \bar{x}\, \neg\theta(\bar{x})$, contradicting $\mathcal{M} \models \exists \bar{x}\, \theta(\bar{x})$. So every finite subset is consistent with $\mathrm{Th}(\mathcal{N})$, as required.

(2) implies (3): Immediate, since κ-regular ultrafilters exist.

(3) implies (1): Suppose $\mathcal{M}$ embeds into $\mathcal{N}^{\mathcal{U}}$. Since $\mathcal{N} \equiv \mathcal{N}^{\mathcal{U}}$ (diagonal embedding), we have $\mathrm{Th}_\forall(\mathcal{N}) \subseteq \mathrm{Th}(\mathcal{N}) = \mathrm{Th}(\mathcal{N}^{\mathcal{U}})$, so $\mathcal{N}^{\mathcal{U}} \models \mathrm{Th}_\forall(\mathcal{N})$. Universal sentences are preserved under passing to substructures (if $\sigma = \forall \bar{x}\, \theta(\bar{x})$ with θ quantifier-free holds in a structure $\mathcal{P}$, it holds in any substructure $\mathcal{Q} \subseteq \mathcal{P}$, since quantifier-free formulae are absolute between $\mathcal{Q}$ and $\mathcal{P}$). Since $\mathcal{M}$ embeds into $\mathcal{N}^{\mathcal{U}}$, $\mathcal{M} \models \mathrm{Th}_\forall(\mathcal{N})$. □

10 Good ultrafilters

Good ultrafilters are "good" in the sense that all ultraproducts with respect to them give rise to saturated models. We shall impose sufficient conditions on $\mathcal{U}$ to achieve this. Fix the following.

- An infinite cardinal κ, an index set I and an ultrafilter $\mathcal{U}$ over I;
- A language $\mathcal{L}$ with $|\mathcal{L}| \leq \kappa$;
- A family of $\mathcal{L}$-structures $\{\mathcal{M}_i : i \in I\}$ and their ultraproduct $\mathcal{N} := \prod_{\mathcal{U}} \mathcal{M}_i$;
- A set of parameters $A \subseteq N$ such that $|A| \leq \kappa$;
- A complete 1-type $p \in S_1(A)$.
  (Sometimes we drop the variables/parameters and just say $\varphi \in p$ instead of $\varphi(x, a) \in p$).

Consider the set of functions $\mathbb{P} = \{f : [p]^{<\omega} \to \mathcal{U}\}$ and $\leq_{\mathbb{P}}$ defined as $g \leq_{\mathbb{P}} f \Leftrightarrow (\forall u \in [p]^{<\omega})(g(u) \subseteq f(u))$. If $g \leq_{\mathbb{P}} f$, then g is called a *refinement* of f.

Example (Łoś' map)**.** The map $\text{Łoś} : [p]^{<\omega} \to \mathcal{U}$ defined below is an example of a function in $\mathbb{P}$.

$$\text{Łoś}(u) := \left\{ i \in I : \mathcal{M}_i \models \exists x \bigwedge_{\varphi \in u} \varphi(x, a(i)) \right\}$$

Note that since p is a type over $A \subseteq N$, it is finitely satisfiable in $\mathcal{N}$. So for each $u \in [p]^{<\omega}$, $\mathcal{N} \models \exists x \bigwedge_{\varphi \in u} \varphi(x)$. So by Łoś' theorem, $\text{Łoś}(u) \in \mathcal{U}$

We impose certain conditions on $f \in \mathbb{P}$:

(a) ***f* refines the Łoś map**: $f \leq_{\mathbb{P}} \text{Łoś}$;

(b) ***f* is locally finite**: $(\forall i \in I)(\{u \in [p]^{<\omega} : i \in f(u)\}$ is finite);

(c) ***f* is multiplicative**: $(\forall u, v \in [p]^{<\omega})(f(u \cup v) = f(u) \cap f(v))$;

(d) ***f* is decreasing**: $(\forall u, v \in [p]^{<\omega})(u \subseteq v \Rightarrow f(v) \subseteq f(u))$.

The functions $f \in \mathbb{P}$ are also sometimes called *concern functions*. Essentially, $f(u)$ tells us which $\mathcal{M}_i$'s should be concerned with realizing the formulae $\varphi \in u$. This perspective provides better descriptions of the conditions above.

(a) ***f* refines the Łoś map** ensures that each $\mathcal{M}_i$ is only concerned with formulae for which it already has existential witnesses. Since the range of f is $\mathcal{U}$, this fact along with Łoś' theorem, will help us realize those formulae in the ultrapower.

(b) ***f* is locally finite** ensure that each $\mathcal{M}_i$ is only concerned with finitely-many $u \in [p]^{<\omega}$.

Before talking about the other conditions, we make a note of the fact that (c) ⇒ (d). To see this, suppose $u \subseteq v$. Then $f(v) = f(u \cup (v \setminus u)) = f(u) \cap f(v \setminus u)$, which implies $f(v) \subseteq f(u)$.

(d) ***f* is decreasing** implies that if some $\mathcal{M}_i$ is concerned with formulae in v, then it must be concerned about all subsets $u \subseteq v$. In particular, this implies that $f(u \cup v) \subseteq f(u)$ and $f(u \cup v) \subseteq f(v)$. Thus, $f(u \cup v) \subseteq f(u) \cap f(v)$.

(c) ***f* is multiplicative** ensures that if $\mathcal{M}_i \models \exists x \bigwedge u(x)$ and $\mathcal{M}_i \models \exists x \bigwedge v(x)$, then $\mathcal{M}_i \models \exists x \left(\bigwedge u(x) \wedge \bigwedge v(x)\right)$.

Lemma 10.1. Let $f \in \mathbb{P}$ satisfy (c). Then f satisfies (b) $\Leftrightarrow (\forall i \in I)(\exists u_i \in [p]^{<\omega})$ such that

$$(\forall v \in [p]^{<\omega})(i \in f(v) \Leftrightarrow v \subseteq u_i)$$

Proof. ($\Rightarrow$) By local finiteness, $\{u \in [p]^{<\omega} : i \in f(u)\}$ is finite. Let $u_i = \bigcup_{i \in f(u)} u \in [p]^{<\omega}$. Now suppose $v \in [p]^{<\omega}$ is such that $i \in f(v)$. Clearly $v \in \{u \in [p]^{<\omega} : i \in f(u)\}$, which implies $v \subseteq u_i$. Conversely, if $v \subseteq u_i$, then $f(u_i) \subseteq f(v)$. By multiplicativity, $f(u_i) = \bigcap_{i \in f(u)} f(u) \subseteq f(v)$. Clearly, $i \in f(u_i) \subseteq f(v)$.

($\Leftarrow$) Suppose for all $i \in I$, such a $u_i \in [p]^{<\omega}$ exists. Then $i \in f(v) \Leftrightarrow v \subseteq u_i$. Since u_i is finite, there are only finitely-many such subsets $v \subseteq u_i$. So f satisfies (b), i.e. is locally finite. □

Before we proceed further, it maybe useful to give an overview of what we are trying to do. The existence of an $f \in \mathbb{P}$ satisfying the conditions (a), (b) and (c) is sufficient to show that $\prod_{\mathcal{U}} \mathcal{M}_i$ is κ^+-saturated. However, for κ-regular ultrafilters, we only get existence of $f \in \mathbb{P}$ satisfying (a), (b) and (d).

The definition of a (κ^+-) good ultrafilter is exactly the missing piece that allows one to refine an $f \in \mathbb{P}$ satisfying (d) to a $g \leq_{\mathbb{P}} f$ that satisfies (c).

Proposition 10.2. $\mathcal{U}$ is κ-regular $\Rightarrow \exists f \in \mathbb{P}$ satisfying (a), (b) and (d)

Proof. Let $p = \{\varphi_\alpha : \alpha < \kappa\}$ and $\mathcal{E} = \{E_\alpha : \alpha < \kappa\}$ be a κ-regularizing set for $\mathcal{U}$. Define $f \in \mathbb{P}$ as follows.

$$f\big(\{\varphi_{\alpha_1}, \ldots, \varphi_{\alpha_n}\}\big) := \text{Łoś}\big(\{\varphi_{\alpha_1}, \ldots, \varphi_{\alpha_n}\}\big) \bigcap \big(E_{\alpha_1} \cap \ldots \cap E_{\alpha_n}\big)$$

It is clear that f satisfies (a) and (d). We show that it satisfies (b), i.e., it is locally finite.

Towards a contradiction, suppose $\exists i \in I$ such that there are infinitely-many $u \in [p]^{<\omega}$ with $i \in f(u)$. Enumerate countably-many of them as $V = \{u_k : k < \omega\}$. Then, $\bigcup_{k<\omega} u_k$ is infinite. Choose an infinite family of formulae $\{\varphi_{\alpha_k} : k < \omega\}$ such that $\varphi_{\alpha_k} \in u_k$ for all $k < \omega$.

$$i \in \bigcap_{k<\omega} f(u_k) \subseteq \bigcap_{k<\omega} f(\{\varphi_{\alpha_k}\}) \subseteq \bigcap_{k<\omega} E_{\alpha_k}$$

This contradicts the κ-regularity of $\mathcal{U}$. □

An $f \in \mathbb{P}$ satisfying (a), (b) and (d) is called a *distribution* for p. The properties (a) and (b) are preserved under taking refinements. The next result shows that if some distribution for p has a multiplicative refinement, i.e. one that satisfies (c), then p is realized in $\mathcal{N} = \prod_{\mathcal{U}} \mathcal{M}_i$.

Proposition 10.3. Let $f \in \mathbb{P}$ satisfy (a), (b) and (c). Then p is realized in $\mathcal{N}$.

Proof. We define $b \in N$ realizing p as follows. For $i \in I$, let u_i be as given by **Lemma 10.1**. Define $b(i) = x \in M_i$ as the witness for $\mathcal{M}_i \models \exists x \bigwedge_{\varphi \in u_i} \varphi(x)$. Such a witness exists, since $\varphi \in u_i \Leftrightarrow \{\varphi\} \subseteq u_i \Leftrightarrow i \in f(\{\varphi\})$, by **Lemma 10.1** and the fact that $f(\{\varphi\}) \subseteq \text{Łoś}(\{\varphi\})$.

Now suppose $\varphi \in p$. By **Lemma 10.1**, $f(\{\varphi\}) = \{i \in I : \varphi \in u_i\}$. For $i \in f(\{\varphi\})$, since $\varphi \in u_i$ and $b(i)$ witnesses $\mathcal{M}_i \models \exists x \bigwedge_{\psi \in u_i} \psi(x)$, we get $\mathcal{M}_i \models \varphi(b(i))$. So $f(\{\varphi\}) \subseteq \{i : \mathcal{M}_i \models \varphi(b(i))\}$, and since $f(\{\varphi\}) \in \mathcal{U}$ and $\mathcal{U}$ is closed upward, $\{i : \mathcal{M}_i \models \varphi(b(i))\} \in \mathcal{U}$. By Łoś' theorem, $\mathcal{N} \models \varphi(b)$. Since $\varphi \in p$ was arbitrary, b realizes p in $\mathcal{N}$. □

We are now ready to define good ultrafilters. Two things need to kept in mind, while making this definition. Firstly, we can only talk about the ultrafilter $\mathcal{U}$ and not the types $p \in S_1(A)$. Secondly, we must ensure that distribution for all types $p \in S_1(A)$ should have a multiplicative refinement.

Definition 10.4 (Good ultrafilter)**.** An ultrafilter $\mathcal{U}$ is κ^+-good if every decreasing (i.e. satisfying (d)) $f : [\kappa]^{<\omega} \to \mathcal{U}$ has a multiplicative refinement (i.e. satisfies (c)).

Proposition 10.5. $\mathcal{U}$ is countably incomplete and κ^+-good $\Rightarrow \mathcal{U}$ is κ-regular.

Proof. Fix $(I_n)_{n<\omega}$ from $\mathcal{U}$ with $I_n \supseteq I_{n+1}$ and $\bigcap_n I_n = \emptyset$. Define $f : [\kappa]^{<\omega} \to \mathcal{U}$ by $f(w) := I_{|w|}$. Clearly f is decreasing (condition (d)): if $w \subseteq w'$ then $|w| \leq |w'|$, so $f(w') = I_{|w'|} \subseteq I_{|w|} = f(w)$. Since $\mathcal{U}$ is κ^+-good, f has a multiplicative refinement $g : [\kappa]^{<\omega} \to \mathcal{U}$.

Let $E_\alpha := g(\{\alpha\})$ for $\alpha < \kappa$; we claim $E := \{E_\alpha : \alpha < \kappa\}$ is a κ-regularizing set for $\mathcal{U}$. Suppose toward a contradiction that $i \in I$ belongs to infinitely many E_α; fix distinct $\alpha_1, \alpha_2, \ldots$ with $i \in E_{\alpha_k}$ for all k, and let $w_n := \{\alpha_1, \ldots, \alpha_n\}$. By multiplicativity, $i \in \bigcap_{k \leq n} E_{\alpha_k} = g(w_n) \subseteq f(w_n) = I_n$ for every n (using that g refines f). So $i \in \bigcap_n I_n = \emptyset$, a contradiction. So $\mathcal{U}$ is κ-regular. □

Corollary 10.6. Let $\mathcal{U}$ be a countably incomplete ultrafilter over I with $|I| = \kappa$. Then $\mathcal{U}$ cannot be κ^{++}-good.

Proof. Suppose toward a contradiction that $\mathcal{U}$ is κ^{++}-good. Since $\mathcal{U}$ is also countably incomplete, the previous proposition (applied with κ^+ in place of κ) gives that $\mathcal{U}$ is κ^+-regular. But κ-regularity requires $\kappa^+ \leq |I| = \kappa$, which is false. Contradiction. □

Theorem 10.7. $\mathcal{L}$ be a language with $|\mathcal{L}| \leq \kappa$ and $\{\mathcal{M}_i : i \in I\}$ be a family of $\mathcal{L}$-structures. If $\mathcal{U}$ is a countably incomplete κ^+-good ultrafilter over I, then $\prod_{\mathcal{U}} \mathcal{M}_i$ is κ^+-saturated.

Proof. Let $\mathcal{N} := \prod_{\mathcal{U}} \mathcal{M}_i$, let $A \subseteq N$ with $|A| \leq \kappa$, and let $p \in S_1(A)$. Since $\mathcal{U}$ is countably incomplete and κ^+-good, $\mathcal{U}$ is κ-regular by the proposition above. Since $|p| \leq |\mathcal{L}_A| \leq |\mathcal{L}| + |A| \leq \kappa$, the earlier proposition (using κ-regularity) gives a distribution f for p, i.e. $f \in \mathbb{P}$ satisfying (a), (b), and (d), which we may view as a decreasing function $[\kappa]^{<\omega} \to \mathcal{U}$ after identifying p with (a subset of) κ via a bijection $|p| \leq \kappa$. Since $\mathcal{U}$ is κ^+-good, f has a multiplicative refinement g. Since refinements preserve properties (a) and (b), g satisfies (a), (b), and (c). By the proposition above, p is realized in $\mathcal{N}$. Since A and p were arbitrary (subject to $|A| \leq \kappa$), $\mathcal{N}$ is κ^+-saturated. □

We have now seen that the κ^+-good ultrafilters are exactly the ultrafilters that yield κ^+-saturated ultraproducts (when the language has size at most κ). We record some variations of this fact, and work towards a converse of the theorem above: that κ^+-saturation of ultraproducts (for a countably incomplete $\mathcal{U}$) forces $\mathcal{U}$ to be κ^+-good.

Suppose $\mathcal{U}$ is κ-regular but *not* κ^+-good, so there is a decreasing $f : [\kappa]^{<\omega} \to \mathcal{U}$ with no multiplicative refinement. Given a set Σ of formulae realized by some $[b]_{\mathcal{U}} \in \mathcal{N}$ via a distribution C, one might hope that $C'(u) := \{i \in I : \mathcal{M}_i \models \bigwedge_{\varphi \in u} \varphi(b(i))\} \cap C(u)$ is a multiplicative refinement of C. It clearly refines C and stays within $\mathcal{U}$ (as $[b]_{\mathcal{U}}$ realizes Σ). To check multiplicativity of C', suppose $i \in C'(u) \cap C'(v)$; the issue is whether $i \in C'(u \cup v)$ requires knowing not just that $\mathcal{M}_i$ satisfies u and v separately at $b(i)$, but that C itself is multiplicative. This motivates the following notion.

Definition 10.8. A distribution $C : [\Sigma]^{<\omega} \to \mathcal{U}$ (for a set Σ of formulae) is called **accurate** if, for every $u \in [\Sigma]^{<\omega}$, whenever $i \in C(\{\varphi\})$ for every $\varphi \in u$, then $i \in C(u)$ if and only if $\mathcal{M}_i \models \exists x \bigwedge_{\varphi \in u} \varphi(x)$.

Proposition 10.9. If Σ is realized in $\mathcal{N}$, then every accurate distribution for Σ has a multiplicative refinement.

Proof. Suppose $[b]_{\mathcal{U}} \in N$ realizes Σ and C is an accurate distribution for Σ. Define $C'(u) := \{i \in I : \mathcal{M}_i \models \bigwedge_{\varphi \in u} \varphi(b(i))\} \cap C(u)$. Since $[b]_{\mathcal{U}}$ realizes Σ, $\{i : \mathcal{M}_i \models \bigwedge_{\varphi \in u} \varphi(b(i))\} \in \mathcal{U}$ (Łoś' theorem), so $C'(u) \in \mathcal{U}$, and clearly C' refines C (hence is again a distribution).

We show C' is multiplicative. Suppose $i \in C'(u) \cap C'(v)$; we must show $i \in C'(u \cup v)$, i.e. $i \in C(u \cup v)$ (the other conjunct, $\mathcal{M}_i \models \bigwedge_{\varphi \in u \cup v} \varphi(b(i))$, follows immediately from $i \in C'(u) \cap C'(v)$). Since $i \in C'(u) \subseteq C(\{\varphi\})$ for each $\varphi \in u$ (as C is antimonotonic and $\{\varphi\} \subseteq u$), and similarly for v, we have $i \in C(\{\varphi\})$ for every $\varphi \in u \cup v$. Since also $\mathcal{M}_i \models \bigwedge_{\varphi \in u \cup v} \varphi(b(i))$ (from $i \in C'(u) \cap C'(v)$), in particular $\mathcal{M}_i \models \exists x \bigwedge_{\varphi \in u \cup v} \varphi(x)$. By accuracy of C (applied to $u \cup v$), $i \in C(u \cup v)$, as desired. □

Accurate distributions exist in any regular ultraproduct:

Lemma 10.10. Suppose $\mathcal{U}$ is κ-regular. Then Σ has an accurate distribution.

Proof. The distribution constructed in the proof of the proposition above (showing κ-regularity gives a distribution satisfying (a), (b), (d)) is accurate. Recall that distribution was

$$f(\{\varphi_{\alpha_1}, \ldots, \varphi_{\alpha_n}\}) := \text{Łoś}(\{\varphi_{\alpha_1}, \ldots, \varphi_{\alpha_n}\}) \cap (E_{\alpha_1} \cap \cdots \cap E_{\alpha_n}),$$

where $\{E_\alpha : \alpha < \kappa\}$ is a κ-regularizing set. To see accuracy, suppose $i \in f(\{\varphi_\alpha\})$ for every $\varphi_\alpha \in u$ and $\mathcal{M}_i \models \exists x \bigwedge_{\varphi \in u} \varphi(x)$; then $i \in \text{Łoś}(u) \cap \bigcap_{\varphi_\alpha \in u} E_\alpha = f(u)$, by definition of the Łoś map and f. Conversely, if $i \in f(u)$ then by definition $i \in \text{Łoś}(u)$, i.e. $\mathcal{M}_i \models \exists x \bigwedge_{\varphi \in u} \varphi(x)$. □

Summarizing the above:

Theorem 10.11. Suppose $\mathcal{U}$ is a κ-regular ultrafilter. Then the following are equivalent:

1. Some distribution of Σ has a multiplicative refinement.
2. Σ is realized in $\mathcal{N}$.
3. Every accurate distribution of Σ has a multiplicative refinement.

Proof. (1) implies (2) is the proposition on realizing p via a multiplicative refinement, applied to Σ. (2) implies (3) is the proposition above. (3) implies (1) follows from the lemma above (accurate distributions exist, and are in particular distributions). □

We are now ready to carry out the key argument establishing the converse of the earlier theorem.

Theorem 10.12. Suppose $\mathcal{U}$ is a κ-regular ultrafilter that is not κ^+-good. Let $\mathcal{M}$ be the structure whose universe is $[\omega]^{<\omega}$ and which possesses a single binary relation $\subseteq$ interpreted as subset. Then $\mathcal{M}^{\mathcal{U}}$ is not κ^+-saturated.

Proof. By the theorem relating regularity and κ^+-saturated elementarily equivalent structures (**Theorem 9.13**), we may replace $\mathcal{M}$ with a κ^+-saturated elementarily equivalent structure $\mathcal{M}'$. Since $\mathcal{U}$ is not κ^+-good, there is a decreasing $f : [\kappa]^{<\omega} \to \mathcal{U}$ with no multiplicative refinement. We use f to produce a set Σ

of formulae with parameters from $(\mathcal{M}')^{\mathcal{U}}$ of size at most κ and an accurate distribution C for Σ with no multiplicative refinement; by the theorem above, this shows Σ is not realized in $(\mathcal{M}')^{\mathcal{U}}$, so $(\mathcal{M}')^{\mathcal{U}}$ is not κ^+-saturated.

Fix $i \in I$ and temporarily call $j < \kappa$ *i-relevant* if $i \in f(\{j\})$. Consider the set $\Gamma_i(y_j)$ of formulae (in the free variables y_j for j i-relevant):

$$\Gamma_i(y_j) := \left\{ \exists x \bigwedge_{l \leq k} x \subseteq y_{j_l} \ : \ i \in f(\{j_1, \ldots, j_k\}) \right\} \cup \left\{ \neg \exists x \bigwedge_{l \leq k} x \subseteq y_{j_l} \ : \ i \notin f(\{j_1, \ldots, j_k\}) \right\}.$$

We leave it to the reader to check that $\Gamma_i(y_j)$ is finitely satisfiable in $\mathcal{M}'$, hence satisfiable in $\mathcal{M}'$ by κ^+-saturation. For $j < \kappa$ that is i-relevant, let $a_j(i) \in M'$ realize Γ_i; for j not i-relevant, take $a_j(i) \in M'$ arbitrary (outside $\mathcal{U}$-almost all i, so negligible).

We now have elements $[a_j]_{\mathcal{U}} \in (M')^{\mathcal{U}}$ for $j < \kappa$. Set $\varphi_j(x) := x \subseteq [a_j]_{\mathcal{U}}$ and $\Sigma := \{\varphi_j(x) : j < \kappa\}$. Let $\{E_j : j < \kappa\}$ be a regularizing subset of $\mathcal{U}$ and define $C : [\Sigma]^{<\omega} \to \mathcal{U}$ by first setting $C(\{\varphi_j\}) := f(\{j\}) \cap E_j$, then extending to all of $[\Sigma]^{<\omega}$ by

$$C(\{\varphi_{j_1}, \ldots, \varphi_{j_k}\}) := \bigcap_{l \leq k} C(\{\varphi_{j_l}\}) \cap \text{Łoś}(\{\varphi_{j_1}, \ldots, \varphi_{j_k}\}) \cap f(\{j_1, \ldots, j_k\}).$$

We leave it to the reader to check that C is an accurate distribution for Σ. Since C refines f, C has no multiplicative refinement taking values in $\mathcal{U}$ (else f would too). Consequently, Σ is not realized in $(\mathcal{M}')^{\mathcal{U}}$, and thus $(\mathcal{M}')^{\mathcal{U}}$, hence $\mathcal{M}^{\mathcal{U}}$, is not κ^+-saturated, as desired. □

We have now seen that the κ^+-good ultrafilters are exactly the ultrafilters that yield κ^+-saturated ultraproducts, when the language has size at most κ. We can also record other equivalent formulations:

Theorem 10.13. Let $\mathcal{U}$ be a countably incomplete ultrafilter on I. The following are equivalent.

1. $\mathcal{U}$ is κ^+-good.
2. For any language $\mathcal{L}$ with $|\mathcal{L}| \leq \kappa$, all $\mathcal{U}$-ultraproducts of $\mathcal{L}$-structures are κ^+-saturated.
3. For any language $\mathcal{L}$ with $|\mathcal{L}| \leq \kappa$, all $\mathcal{U}$-ultraproducts of $\mathcal{L}$-structures are κ^+-universal.
4. For any language $\mathcal{L}$ with $|\mathcal{L}| \leq \kappa$, all $\mathcal{U}$-ultrapowers of $\mathcal{L}$-structures are κ^+-saturated.

Proof. (1) implies (2) is the theorem proved above. (2) implies (3) follows from the fact (proved in **Proposition 6.6**) that κ-saturation implies κ^+-universality.

(3) implies (4): Suppose $|\mathcal{L}| \leq \kappa$ and $\mathcal{M}$ is an $\mathcal{L}$-structure; we wish to show $\mathcal{M}^{\mathcal{U}}$ is κ^+-saturated. By **Proposition 6.7** relating saturation to universality of parametrized reducts, it suffices to show $(\mathcal{M}^{\mathcal{U}}; a)$ is κ^+-universal for every $a \in (M^{\mathcal{U}})^\kappa$. This follows from (3), the observation that $(\mathcal{M}^{\mathcal{U}}; a) = \prod_{\mathcal{U}}(\mathcal{M}; a(i))$, and the fact that $|\mathcal{L}(\kappa)| \leq \kappa$.

(4) implies (1): Suppose (4) holds and $\mathcal{U}$ is not κ^+-good. Since κ^+-saturation implies κ^+-universality, the regularity-universality theorem (**Theorem 9.10**) shows $\mathcal{U}$ is κ-regular. We thus obtain a contradiction using the previous theorem, which shows some $\mathcal{U}$-ultrapower is not κ^+-saturated. □

Corollary 10.14. Every non-principal ultrafilter on ω is good.

Proof. This follows immediately from **Theorem 7.8** together with the previous theorem, taking $\kappa := \omega$. □

Remark. One can also prove the previous corollary directly from the definition of $\aleph_1$-good, without appealing to the equivalence above; we omit the details, as they amount to a careful direct construction of multiplicative refinements over ω.

Proposition 10.15. Suppose $\mathcal{V}$ is κ-regular. Then, for any ultrafilter $\mathcal{U}$, $\mathcal{U} \times \mathcal{V}$ is κ^+-good if and only if $\mathcal{V}$ is κ^+-good.

Proof. Recall from Section 5 that $\prod_{\mathcal{U}\times\mathcal{V}} \mathcal{M} \cong \prod_{\mathcal{V}} (\prod_{\mathcal{U}} \mathcal{M})$ for every $\mathcal{L}$-structure $\mathcal{M}$. By the theorem above (applied to ultrapowers, condition (4)), it suffices to work with ultrapowers: $\mathcal{U} \times \mathcal{V}$ is κ^+-good if and only if $\mathcal{M}^{\mathcal{U}\times\mathcal{V}}$ is κ^+-saturated for every $\mathcal{L}$-structure $\mathcal{M}$ with $|\mathcal{L}| \leq \kappa$, if and only if $(\mathcal{M}^{\mathcal{U}})^{\mathcal{V}}$ is κ^+-saturated for every such $\mathcal{M}$.

($\Leftarrow$) If $\mathcal{V}$ is κ^+-good, then $\mathcal{N}^{\mathcal{V}}$ is κ^+-saturated for every $\mathcal{L}$-structure $\mathcal{N}$ with $|\mathcal{L}| \leq \kappa$ (condition (4) for $\mathcal{V}$); taking $\mathcal{N} := \mathcal{M}^{\mathcal{U}}$ gives $(\mathcal{M}^{\mathcal{U}})^{\mathcal{V}}$ κ^+-saturated for every $\mathcal{M}$, so $\mathcal{U} \times \mathcal{V}$ is κ^+-good.

($\Rightarrow$) Suppose $\mathcal{U} \times \mathcal{V}$ is κ^+-good, and fix an $\mathcal{L}$-structure $\mathcal{N}$ with $|\mathcal{L}| \leq \kappa$. Let $\mathcal{M} := \mathcal{N}^{\mathcal{U}}$; since $\mathcal{N} \equiv \mathcal{M}$ (diagonal embedding), and $(\mathcal{M})^{\mathcal{V}} = (\mathcal{N}^{\mathcal{U}})^{\mathcal{V}} \cong \mathcal{N}^{\mathcal{U}\times\mathcal{V}}$ is κ^+-saturated (as $\mathcal{U} \times \mathcal{V}$ is κ^+-good), the theorem of Keisler on regular ultrafilters (**Theorem 9.13**) gives that $\mathcal{N}^{\mathcal{V}}$ is also κ^+-saturated. Since $\mathcal{N}$ was arbitrary, $\mathcal{V}$ is κ^+-good. □

Corollary 10.16. Suppose $|\mathcal{L}| \leq \kappa$ and that $\mathcal{M}$ and $\mathcal{N}$ are $\mathcal{L}$-structures with $|M|, |N| \leq \kappa^+$. Suppose $2^\kappa = \kappa^+$. Suppose $\mathcal{U}$ is a good ultrafilter on κ. Then $\mathcal{M} \equiv \mathcal{N}$ if and only if $\mathcal{M}^{\mathcal{U}} \cong \mathcal{N}^{\mathcal{U}}$.

Proof. The only direction that needs proving is the forward direction. If $\mathcal{M} \equiv \mathcal{N}$, then $\mathcal{M}^{\mathcal{U}}$ and $\mathcal{N}^{\mathcal{U}}$ are elementarily equivalent κ^+-saturated models of size 2^κ, by the cardinality formula for regular ultrapowers (**Theorem 9.9**) and the theorem above (as $\mathcal{U}$ is good). Since we are assuming $2^\kappa = \kappa^+$, both have size κ^+, so by the saturated-isomorphism theorem (**Theorem 6.4**), $\mathcal{M}^{\mathcal{U}} \cong \mathcal{N}^{\mathcal{U}}$. □

Corollary 10.17. Assume the Generalized Continuum Hypothesis. Then for any language $\mathcal{L}$ and any $\mathcal{L}$-structures $\mathcal{M}$ and $\mathcal{N}$, $\mathcal{M} \equiv \mathcal{N}$ if and only if $\mathcal{M}$ and $\mathcal{N}$ have isomorphic ultrapowers.

Proof. ($\Rightarrow$) Let $\kappa := \max(|M|, |N|, |\mathcal{L}|, \aleph_0)$; then $|M|, |N| \leq \kappa \leq \kappa^+$ and $|\mathcal{L}| \leq \kappa$. Under GCH, $2^\kappa = \kappa^+$, so the previous corollary applies (using the existence of good ultrafilters on κ, to be established in **Theorem 11.6**) to give $\mathcal{M}^{\mathcal{U}} \cong \mathcal{N}^{\mathcal{U}}$ for a good ultrafilter $\mathcal{U}$ on κ.

($\Leftarrow$) Ultrapowers are elementary extensions, so isomorphic ultrapowers of $\mathcal{M}$ and $\mathcal{N}$ give $\mathcal{M} \equiv \mathcal{M}^{\mathcal{U}} \cong \mathcal{N}^{\mathcal{V}} \equiv \mathcal{N}$. □

11 Existence of good ultrafilters

We now prove the existence of good ultrafilters. The rough idea is to construct an increasing sequence of filters that eventually become an ultrafilter (we take care of this at "even" steps of the construction) and which satisfies the goodness property (which we take care of at "odd" steps). As we go along, in order to ensure that we can continue, we need to insist that there are "many" possible extensions of our filter; the notion of many possible extensions is made precise by the notion of consistent pair defined below.

Definition 11.1. Fix a cardinal κ. Let Π be a non-empty set of partitions of κ such that each partition in Π has κ many (non-empty) pieces. Let $\mathcal{F}$ be a filter on κ. We say that $(\Pi, \mathcal{F})$ is **consistent** if, given any $X \in \mathcal{F}$, $P_1, \ldots, P_n \in \Pi$, and $X_1 \in P_1, \ldots, X_n \in P_n$, we have that $X \cap \bigcap_{i=1}^n X_i \neq \emptyset$.

We work towards proving the existence of good ultrafilters. Our first lemma is a set-theoretic one.

Lemma 11.2. Suppose κ is an infinite cardinal and, for each $\alpha < \kappa$, Y_α is a set of cardinality κ. Then, for each $\alpha < \kappa$, there is $Z_\alpha \subseteq Y_\alpha$ of cardinality κ such that $Z_\alpha \cap Z_\beta = \emptyset$ for $\alpha \neq \beta$.

Proof. For each $\alpha \leq \kappa$, let $T_\alpha := \{(\beta, \gamma) : \beta \leq \gamma < \alpha\}$; thinking of the "discrete grid" $\alpha \times \alpha$, T_α is the set of points on or above the diagonal. For $\beta < \alpha < \kappa$, let $T_{\alpha,\beta} := T_\alpha \cap (\{\beta\} \times \alpha)$, the vertical slice of T_α above β. We define, by induction on $\alpha < \kappa$, an increasing chain of injective functions f_α with domain T_α such that $f(T_{\alpha,\beta}) \subseteq Y_\beta$ for each $\beta < \alpha$. Indeed, suppose f_α has been constructed. Note $T_{\alpha+1}$ simply adds one more point to each vertical slice of T_α; since $|T_\alpha| < \kappa$ and each $|Y_\beta| = \kappa$, it is clear how to extend f_α to an injective function $f_{\alpha+1}$ on $T_{\alpha+1}$. Taking unions at limits preserves the requirements of the recursion.

We now set $f := \bigcup_{\alpha<\kappa} f_\alpha$. Note f is injective and $f(T_{\kappa,\beta}) \subseteq Y_\beta$ for each $\beta < \kappa$. Since $|T_{\kappa,\beta}| = \kappa$ and f is injective, setting $Z_\beta := f(T_{\kappa,\beta})$ yields the desired sets. □

Lemma 11.3. Suppose $\mathcal{F}$ is a uniform filter on κ generated by $E \subseteq \mathcal{F}$, $|E| \leq \kappa$. Then there is a set Π of partitions of κ with $|\Pi| = 2^\kappa$ such that $(\Pi, \mathcal{F})$ is consistent.

Proof. Without loss of generality, E is closed under finite intersections. Let $(J_\alpha : \alpha < \kappa)$ enumerate E; since $\mathcal{F}$ is uniform, each J_α has cardinality κ. By the previous lemma, there are $I_\alpha \subseteq J_\alpha$ such that each I_α has cardinality κ and $I_\alpha \cap I_\beta = \emptyset$ for $\alpha \neq \beta$.

Let $(f_\gamma)_{\gamma<\kappa}$ be an enumeration of all functions $f : \mathcal{P}(u) \to \kappa$, where $u \subseteq \kappa$ is finite (write u_γ for the domain-defining set of f_γ). Without loss of generality, for every $\alpha < \kappa$, $\{f_\gamma : \gamma \in I_\alpha\}$ itself is an enumeration of the set of all such functions (recall each I_α has cardinality κ).

For each $X \subseteq \kappa$, define $f_X : \kappa \to \kappa$ by $f_X(\gamma) := f_\gamma(X \cap u_\gamma)$ if $\gamma \in I_\alpha$ for some $\alpha < \kappa$, and $f_X(\gamma) := 0$ otherwise. Let $P_X := \{f_X^{-1}(\delta) : \delta < \kappa\}$; this is a partition of κ into exactly κ many pieces.

We first show $P_X \neq P_Y$ for distinct $X, Y \subseteq \kappa$ (whence there are 2^κ many such partitions, as $X \mapsto P_X$ is then injective). Suppose, without loss of generality, $\eta \in X \setminus Y$. Let $f : \mathcal{P}(\{\eta\}) \to \kappa$ be given by $f(\{\eta\}) = 0$ and $f(\emptyset) = 1$. Fix $\alpha < \kappa$ and take $\gamma \in I_\alpha$ with $f_\gamma = f$. Then $f_X(\gamma) = f_\gamma(X \cap u_\gamma) = f_\gamma(\{\eta\}) = 0$ while $f_Y(\gamma) = f_\gamma(Y \cap u_\gamma) = f_\gamma(\emptyset) = 1$. So $P_X \neq P_Y$.

Let $\Pi := \{P_X : X \subseteq \kappa\}$. We finish by showing $(\Pi, \mathcal{F})$ is consistent. It suffices to show $J_\alpha \cap f_{X_1}^{-1}(\delta_1) \cap \cdots \cap f_{X_n}^{-1}(\delta_n) \neq \emptyset$ for any $\alpha, \delta_1, \ldots, \delta_n < \kappa$ and distinct $X_1, \ldots, X_n \subseteq \kappa$. Take finite $u \subseteq \kappa$ such that $X_i \cap u \neq X_j \cap u$ for distinct i, j. Define $f : \mathcal{P}(u) \to \kappa$ with $f(X_i \cap u) = \delta_i$ for each i (arbitrary elsewhere). Take $\gamma \in I_\alpha$ with $f_\gamma = f$. Then, for each i, $f_{X_i}(\gamma) = f_\gamma(X_i \cap u_\gamma) = f_\gamma(X_i \cap u) = \delta_i$, so γ witnesses the desired non-empty intersection. □

Lemma 11.4. Suppose $(\Pi, \mathcal{F})$ is consistent and $J \subseteq \kappa$. Then either $(\Pi, \langle \mathcal{F}, \{J\}\rangle)$ is consistent, or else there is a cofinite $\Pi' \subseteq \Pi$ for which $(\Pi', \langle \mathcal{F}, \{\kappa \setminus J\}\rangle)$ is consistent.

Proof. Suppose $(\Pi, \langle \mathcal{F}, \{J\}\rangle)$ is not consistent. By definition, there is $X \in \mathcal{F}$, distinct $P_1, \ldots, P_n \in \Pi$, and $X_i \in P_i$ $(i \leq n)$ for which $X \cap J \cap \bigcap_{i=1}^n X_i = \emptyset$. Let $\Pi' := \Pi \setminus \{P_1, \ldots, P_n\}$, a cofinite subset of Π. We claim $(\Pi', \langle \mathcal{F}, \{\kappa \setminus J\}\rangle)$ is consistent. Fix $Y \in \mathcal{F}$, distinct $Q_1, \ldots, Q_m \in \Pi'$, and $X'_j \in Q_j$ $(j \leq m)$. Since $(\Pi, \mathcal{F})$ is consistent, there is $x \in X \cap Y \cap \bigcap_{i=1}^n X_i \cap \bigcap_{j=1}^m X'_j$. By the displayed equation above, $x \notin J$, so $(\kappa \setminus J) \cap Y \cap \bigcap_{j=1}^m X'_j \neq \emptyset$, as desired. □

Lemma 11.5. Suppose $(\Pi, \mathcal{F})$ is consistent, $p : [\kappa]^{<\omega} \to \mathcal{F}$ is antimonotonic, and $P \in \Pi$. Then there is a filter $\mathcal{F}'$ extending $\mathcal{F}$ such that $(\Pi \setminus \{P\}, \mathcal{F}')$ is consistent, and a multiplicative refinement $q : [\kappa]^{<\omega} \to \mathcal{F}'$ of p.

Proof. Let $(X_\delta)_{\delta<\kappa}$ be an injective enumeration of P and $(t_\delta)_{\delta<\kappa}$ an enumeration of $[\kappa]^{<\omega}$. For $\delta < \kappa$, define $q_\delta : [\kappa]^{<\omega} \to \mathcal{P}(\kappa)$ by $q_\delta(u) := p(t_\delta) \cap X_\delta$ if $u \subseteq t_\delta$, and $q_\delta(u) := \emptyset$ otherwise. Note: $q_\delta(u) \subseteq p(t_\delta)$; $q_\delta(u) \neq \emptyset$ whenever $u \subseteq t_\delta$ (by consistency of $(\Pi, \mathcal{F})$, as $p(t_\delta) \in \mathcal{F}$ and $X_\delta \in P \in \Pi$); and q_δ is multiplicative: if

$u \not\subseteq t_\delta$ or $v \not\subseteq t_\delta$, then also $u \cup v \not\subseteq t_\delta$ (as $u, v \subseteq u \cup v$), so $q_\delta(u \cup v) = \emptyset$; moreover one of $q_\delta(u), q_\delta(v)$ is already $\emptyset$ in this case, so $q_\delta(u) \cap q_\delta(v) = \emptyset = q_\delta(u \cup v)$ too. Otherwise $u, v \subseteq t_\delta$, whence $u \cup v \subseteq t_\delta$, and $q_\delta(u \cup v) = q_\delta(u) = q_\delta(v) = p(t_\delta) \cap X_\delta$.

Define $q : [\kappa]^{<\omega} \to \mathcal{P}(\kappa)$ by $q(u) := \bigcup_{\delta<\kappa} q_\delta(u)$. First, q refines p: if $u \subseteq t_\delta$, then $q_\delta(u) = p(t_\delta) \cap X_\delta \subseteq p(u) \cap X_\delta$ by antimonotonicity of p; it follows that $q(u) \subseteq p(u)$. Next, q is multiplicative, immediately from the fact that each q_δ is multiplicative and the X_δ's are disjoint.

Let J denote the range of q and $\mathcal{F}' := \langle \mathcal{F}, J \rangle$. We conclude by showing $(\Pi \setminus \{P\}, \mathcal{F}')$ is consistent. Take $X \in \mathcal{F}$, $u \in [\kappa]^{<\omega}$, distinct $P_1, \ldots, P_n \in \Pi \setminus \{P\}$, and $X_i \in P_i$; we show $X \cap q(u) \cap \bigcap_{i=1}^n X_i \neq \emptyset$ (an arbitrary element of $\mathcal{F}'$ is of the form $X \cap q(u)$, since q is multiplicative). Take δ with $u = t_\delta$; then $q(u) \supseteq q_\delta(u) = p(t_\delta) \cap X_\delta$. Since $(\Pi, \mathcal{F})$ is consistent and $p(t_\delta) \in \mathcal{F}$, $X \cap p(t_\delta) \cap X_\delta \cap \bigcap_{i=1}^n X_i \neq \emptyset$, as desired. □

We now have all the necessary ingredients to prove:

Theorem 11.6. For each κ, there is a countably incomplete κ^+-good ultrafilter on κ.

Proof. Let $(X_\alpha)_{\alpha<2^\kappa}$ enumerate $\mathcal{P}(\kappa)$ and let $(f_\alpha)_{\alpha<2^\kappa}$ enumerate all antimonotonic functions $[\kappa]^{<\omega} \to \mathcal{P}(\kappa)$, with each such function enumerated 2^κ many times. Let F_0 be the filter on κ generated by $E_0 := \{\kappa \setminus \beta : \beta < \kappa\}$, a uniform filter generated by κ many sets; by the lemma above, fix Π_0 with $|\Pi_0| = 2^\kappa$ such that (Π_0, F_0) is consistent.

We define, by transfinite recursion, a sequence of consistent pairs (Π_α, F_α) for $\alpha < 2^\kappa$ satisfying:

1. Each Π_α has cardinality 2^κ.
2. If $\alpha \leq \beta < 2^\kappa$, then $\Pi_\alpha \supseteq \Pi_\beta$ and $F_\alpha \subseteq F_\beta$.
3. If $\alpha < 2^\kappa$ is a limit ordinal, then $\Pi_\alpha := \bigcap_{\beta<\alpha} \Pi_\beta$ and $F_\alpha := \bigcup_{\beta<\alpha} F_\beta$.
4. Each $\Pi_\alpha \setminus \Pi_{\alpha+1}$ is finite.
5. Either $X_\alpha \in F_{\alpha+1}$ or $\kappa \setminus X_\alpha \in F_{\alpha+1}$.
6. If the range of f_α is contained in F_α, then there is a multiplicative refinement $g : [\kappa]^{<\omega} \to F_{\alpha+1}$ of f_α.

Once this is completed, set $\mathcal{U} := \bigcup_{\alpha<2^\kappa} F_\alpha$. By (5), $\mathcal{U}$ is an ultrafilter on κ. $\mathcal{U}$ is countably incomplete since it is uniform (as F_0 already is, by the regularizing sets E_0) and uniform ultrafilters on infinite sets are countably incomplete (regular ultrafilters are countably incomplete, and the construction here is designed to realize a κ-regular, hence countably incomplete, ultrafilter – see also the discussion of regular ultrafilters). By (6), $\mathcal{U}$ is κ^+-good: given antimonotonic $f : [\kappa]^{<\omega} \to \mathcal{U}$, since $\mathrm{cof}(2^\kappa) > \kappa$, there is $\alpha < 2^\kappa$ such that the range of f is contained in F_α; since every antimonotonic function is enumerated 2^κ many times, there is $\beta > \alpha$ with $f = f_\beta$. It follows that f has a multiplicative refinement taking values in $F_{\beta+1} \subseteq \mathcal{U}$, as desired.

We now carry out the recursion; item (4) ensures that the resulting Π_α retain cardinality 2^κ at each stage. Item (3) tells us what to do at limit ordinals.

For the successor step, suppose (Π_β, F_β) have been constructed for all $\beta \leq \alpha$. By the lemma above, there is $\Pi' \subseteq \Pi_\alpha$ with $\Pi_\alpha \setminus \Pi'$ finite and $F' \supseteq F_\alpha$ such that (Π', F') is consistent and either $X_\alpha \in F'$ or $\kappa \setminus X_\alpha \in F'$. If the range of f_α is not contained in F', set $\Pi_{\alpha+1} := \Pi'$ and $F_{\alpha+1} := F'$. Otherwise, by the lemma above, there is $\Pi_{\alpha+1} \subseteq \Pi'$ with $\Pi' \setminus \Pi_{\alpha+1}$ finite (a singleton, even) and $F_{\alpha+1} \supseteq F'$ such that $(\Pi_{\alpha+1}, F_{\alpha+1})$ is consistent and f_α has a multiplicative refinement with range included in $F_{\alpha+1}$. This concludes the recursion, and the proof. □

We end with the two promised strengthenings of the existence theorem.

Corollary 11.7. Suppose $E \subseteq \mathcal{P}(\kappa)$ is such that $|E| \leq \kappa$, each element of E has cardinality κ, and E is closed under finite intersections. Then there is a κ^+-good ultrafilter on κ containing E.

Proof. Since E is closed under finite intersections and each of its elements has cardinality κ (in particular is non-empty), E has the finite intersection property, so the filter F_E generated by E is proper. Moreover F_E is uniform: every element of F_E contains some finite intersection of elements of E, which (as E is closed under finite intersections) is itself an element of E of cardinality κ, so every superset also has cardinality κ. So F_E is a uniform filter on κ generated by E, $|E| \leq \kappa$.

By the lemma above, fix Π_0 with $|\Pi_0| = 2^\kappa$ such that (Π_0, F_E) is consistent. Running exactly the transfinite recursion in the proof of the Existence Theorem above, but starting from (Π_0, F_E) in place of (Π_0, F_0), produces a countably incomplete κ^+-good ultrafilter $\mathcal{U} := \bigcup_{\alpha<2^\kappa} F_\alpha \supseteq F_0 = F_E \supseteq E$, as desired. □

Corollary 11.8. There are 2^{2^κ} many κ^+-good ultrafilters on κ.

Proof. Since every ultrafilter on κ is a subset of $\mathcal{P}(\mathcal{P}(\kappa))$, there are at most 2^{2^κ} ultrafilters on κ in total (indeed exactly 2^{2^κ}, as shown earlier), so it suffices to exhibit 2^{2^κ} many κ^+-good ones.

By the construction used to count ultrafilters earlier (applied with $X := \kappa$), there is an independent family $\{b_s : s \subseteq \kappa\}$ of subsets of a set Y with $|Y| = \kappa$; fixing a bijection between Y and κ, we may take $\{b_s : s \subseteq \kappa\}$ to be an independent family of subsets of κ itself. Fix a function $h : \mathcal{P}(\kappa) \to \{0, 1\}$. We sketch how to produce a κ^+-good ultrafilter $\mathcal{U}_h$ with $b_s \in \mathcal{U}_h \Leftrightarrow h(s) = 1$ for every $s \subseteq \kappa$; since distinct h, h' then yield distinct ultrafilters $\mathcal{U}_h \neq \mathcal{U}_{h'}$ (witnessed by any s with $h(s) \neq h'(s)$), and there are $2^{|\mathcal{P}(\kappa)|} = 2^{2^\kappa}$ many functions h, this gives the desired count.

Run the transfinite recursion from the Existence Theorem, but re-index the enumeration $(X_\alpha)_{\alpha<2^\kappa}$ of $\mathcal{P}(\kappa)$ so that it includes $(b_s)_{s\subseteq\kappa}$ as a sub-enumeration (possible since $\{b_s : s \subseteq \kappa\}$ has cardinality 2^κ, being independent, hence pairwise distinct). At the stage of the recursion handling $X_\alpha = b_s$, instead of invoking the consistency lemma to decide *either* b_s *or* $\kappa \setminus b_s$ (whichever happens to be consistent with what has been built so far), we use the independence of $\{b_s : s \subseteq \kappa\}$ to show that the choice prescribed by $h(s)$ is *always* available: at every stage, the filter built so far is generated by κ-many sets together with finitely many of the $b_{s'}$'s and their complements (chosen consistently with h at earlier stages), and independence of the family $\{b_s\}$ ensures that adjoining whichever of $b_s, \kappa \setminus b_s$ is dictated by $h(s)$ preserves the finite intersection property needed to continue the construction via the consistency lemma, in the direction prescribed by h, rather than being forced to the other side. We omit the routine (if notationally heavy) verification that this modified recursion can be carried out at every stage. Granting this, the resulting $\mathcal{U}_h$ is, exactly as in the Existence Theorem, a countably incomplete κ^+-good ultrafilter on κ with $b_s \in \mathcal{U}_h \Leftrightarrow h(s) = 1$ for every $s \subseteq \kappa$, as required. □

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